



Machine Learning: INN Hotel

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Problem Statement

Business Context

A significant number of hotel bookings are called off due to cancellations or no-shows. The typical reasons for cancellations include change of plans, scheduling conflicts, etc. This is often made easier by the option to do so free of charge or preferably at a low cost, which is beneficial to hotel guests, but it is a less desirable and possibly revenue-diminishing factor for hotels to deal with. Such losses are particularly high on last-minute cancellations. The new technologies involving online booking channels have dramatically changed customers' booking possibilities and behavior. This adds a further dimension to the challenge of how hotels handle cancellations, which are no longer limited to traditional booking and guest characteristics.

The cancellation of bookings impacts a hotel on various fronts:

1. Loss of resources (revenue) when the hotel cannot resell the room.
2. Additional costs of distribution channels by increasing commissions or paying for publicity to help sell these rooms.
3. Lowering prices last minute, so the hotel can resell a room, resulting in a reduction of the profit margin.
4. Human resources to make arrangements for the guests.

Objective

The increasing number of cancellations calls for a Machine Learning-based solution that can help in predicting which booking is likely to be canceled. INN Hotels Group has a chain of hotels in Portugal, and they are facing problems with the high number of booking cancellations and have reached out to your firm for data-driven solutions. As a data scientist have to analyze the data provided to find which factors have a high influence on booking cancellations, build a predictive model that can predict which bookings are going to be canceled in advance, and help in formulating profitable policies for cancellations and refunds.

Data Description

The data contains the different attributes of customers' booking details. The detailed data dictionary is given below.

- Booking_ID: the unique identifier of each booking
- no_of_adults: Number of adults
- no_of_children: Number of Children
- no_of_weekend_nights: Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel
- no_of_week_nights: Number of weeknights (Monday to Friday) the guest stayed or booked to stay at the hotel
- type_of_meal_plan: Type of meal plan booked by the customer:
 - Not Selected – No meal plan selected
 - Meal Plan 1 – Breakfast
 - Meal Plan 2 – Half board (breakfast and one other meal)
 - Meal Plan 3 – Full board (breakfast, lunch, and dinner)
- required_car_parking_space: Does the customer require a car parking space? (0 - No, 1- Yes)

- room_type_reserved: Type of room reserved by the customer. The values are ciphered (encoded) by INN Hotels Group
- lead_time: Number of days between the date of booking and the arrival date
- arrival_year: Year of arrival date
- arrival_month: Month of arrival date
- arrival_date: Date of the month
- market_segment_type: Market segment designation.
- repeated_guest: Is the customer a repeated guest? (0 - No, 1- Yes)
- no_of_previous_cancellations: Number of previous bookings that were canceled * by the customer prior to the current booking
- no_of_previous_bookings_not_canceled: Number of previous bookings not canceled by the customer prior to the current booking
- avg_price_per_room: Average price per day of the reservation; prices of the rooms are dynamic. (in euros)
- no_of_special_requests: Total number of special requests made by the customer (e.g. high floor, view from the room, etc)
- booking_status: Flag indicating if the booking was canceled or not.

Data Overview

The first steps in exploring any dataset typically involve:

- Viewing the first few rows to confirm the dataset has loaded correctly.
- Checking the number of rows and columns to understand its size.
- Reviewing the data types of each column to ensure values are stored in the desired format and match expectations.
- Examining the statistical summary to gain insights into the numerical columns.

Displaying the first few rows of the dataset

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space	room_type_reserved	lead_time	arrival_year	arrival_month	arrival_date	market_segment_type	repeated_guest
0	INN00001	2	0	1	2	Meal Plan 1	0	Room_Type 1	224	2017	10	2	Offline	0
1	INN00002	2	0	2	3	Not Selected	0	Room_Type 1	5	2018	11	6	Online	0
2	INN00003	1	0	2	1	Meal Plan 1	0	Room_Type 1	1	2018	2	28	Online	0
3	INN00004	2	0	0	2	Meal Plan 1	0	Room_Type 1	211	2018	5	20	Online	0
4	INN00005	2	0	1	1	Not Selected	0	Room_Type 1	48	2018	4	11	Online	0

Figure 1: First Few rows

Checking the shape of the dataset

The dataset has 36275 rows and 18 columns

Checking the data types of the columns for the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36275 entries, 0 to 36274
Data columns (total 19 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Booking_ID                               36275 non-null  object
1   no_of_adults                             36275 non-null  int64
2   no_of_children                           36275 non-null  int64
3   no_of_weekend_nights                     36275 non-null  int64
4   no_of_week_nights                       36275 non-null  int64
5   type_of_meal_plan                        36275 non-null  object
6   required_car_parking_space               36275 non-null  int64
7   room_type_reserved                       36275 non-null  object
8   lead_time                               36275 non-null  int64
9   arrival_year                             36275 non-null  int64
10  arrival_month                            36275 non-null  int64
11  arrival_date                             36275 non-null  int64
12  market_segment_type                     36275 non-null  object
13  repeated_guest                           36275 non-null  int64
14  no_of_previous_cancellations             36275 non-null  int64
15  no_of_previous_bookings_not_canceled     36275 non-null  int64
16  avg_price_per_room                       36275 non-null  float64
17  no_of_special_requests                   36275 non-null  int64
18  booking_status                           36275 non-null  object
dtypes: float64(1), int64(13), object(5)
memory usage: 5.3+ MB
```

Figure 2: Data types of the columns

Dataset Overview

- Total records: 36,275 bookings
- Columns: 19, covering booking details, guest info, and booking outcomes
- Target variable: booking_status (likely to indicate if the booking was canceled or not)

Key Categories of Features

Guest Details

no_of_adults, no_of_children, repeated_guest

Booking Details

lead_time, arrival_year/month/date, no_of_weekend_nights, no_of_week_nights, type_of_meal_plan, room_type_reserved

Booking History

no_of_previous_cancellations, no_of_previous_bookings_not_canceled

Operational Details

required_car_parking_space, avg_price_per_room, no_of_special_requests

Market Segment

market_segment_type

Potential Insights to Explore

- Cancellation behavior: Which features influence whether a booking gets canceled (booking_status)?
- Lead time patterns: Do longer lead times correlate with higher cancellations?
- Guest type: Are repeated guests less likely to cancel?
- Room and meal preferences: Which combinations are popular and more stable?
- Seasonality: Month/weekday trends for booking and cancellations.
- Revenue implications: Average price per room vs. cancellation rates.

Statistical summary of the dataset

	count	mean	std	min	25%	50%	75%	max
no_of_adults	36275.0	1.844962	0.518715	0.0	2.0	2.00	2.0	4.0
no_of_children	36275.0	0.105279	0.402648	0.0	0.0	0.00	0.0	10.0
no_of_weekend_nights	36275.0	0.810724	0.870644	0.0	0.0	1.00	2.0	7.0
no_of_week_nights	36275.0	2.204300	1.410905	0.0	1.0	2.00	3.0	17.0
required_car_parking_space	36275.0	0.030986	0.173281	0.0	0.0	0.00	0.0	1.0
lead_time	36275.0	85.232557	85.930817	0.0	17.0	57.00	126.0	443.0
arrival_year	36275.0	2017.820427	0.383836	2017.0	2018.0	2018.00	2018.0	2018.0
arrival_month	36275.0	7.423653	3.069894	1.0	5.0	8.00	10.0	12.0
arrival_date	36275.0	15.596995	8.740447	1.0	8.0	16.00	23.0	31.0
repeated_guest	36275.0	0.025637	0.158053	0.0	0.0	0.00	0.0	1.0
no_of_previous_cancellations	36275.0	0.023349	0.368331	0.0	0.0	0.00	0.0	13.0
no_of_previous_bookings_not_canceled	36275.0	0.153411	1.754171	0.0	0.0	0.00	0.0	58.0
avg_price_per_room	36275.0	103.423539	35.089424	0.0	80.3	99.45	120.0	540.0
no_of_special_requests	36275.0	0.619655	0.786236	0.0	0.0	0.00	1.0	5.0

Figure 3: Statistical summary of the dataset

1. Guest Composition

Adults per booking: On average ~1.84 adults; most bookings are for 2 adults.
Children per booking: Very few children (~0.1 on average); 75% of bookings have no children.
Observation: Majority of bookings are small adult groups, likely couples or solo travelers.

2. Stay Duration

Weekend nights: Average ~0.81 nights; 50% of bookings are for 1 weekend night, maximum 7 nights.
Week nights: Average ~2.2 nights; 50% of bookings are for 2 nights, maximum 17 nights.
Observation: Most bookings are short stays (2–3 nights); long stays are rare.

3. Lead Time (Booking in Advance)

Lead time: Average ~85 days; median 57 days; maximum 443 days.
Observation: A significant number of bookings are made well in advance, but there is a wide variation. Long lead times may correlate with cancellations (potential risk).

4. Parking Requirements

Only ~3% of bookings require a parking space.
Observation: Parking is a minor consideration for most guests.

5. Guest Loyalty & Booking History

Repeated guests are only ~2.5% of total bookings.

Most guests have no previous cancellations; only a few have a history of cancellations.

Observation: The majority are first-time guests. Loyal guests are rare, which might influence marketing strategies.

6. Pricing

Average room price: ₹103.42 per night; 75% of bookings are below ₹120, but the max price is ₹540.

Observation: Most bookings are in the mid-price range, with a few high-end bookings that could significantly impact revenue.

7. Special Requests

Average ~0.62 per booking; 50% of bookings have no special requests.

Observation: Guests rarely make special requests, simplifying operational management.

8. Seasonality & Arrival Patterns

Arrival months: Concentrated between May–October (25–75% percentile).

Arrival year: Mostly 2018; minor data from 2017.

Observation: There is seasonal demand, which can guide staffing and pricing strategies.

	count	unique	top	freq
Booking_ID	36275	36275	INN36275	1
type_of_meal_plan	36275	4	Meal Plan 1	27835
room_type_reserved	36275	7	Room_Type 1	28130
market_segment_type	36275	5	Online	23214
booking_status	36275	2	Not_Canceled	24390

Figure 4: Category Summary

1. Booking IDs

Unique bookings: All 36,275 entries are unique.

Observation: Each row represents a distinct booking, so there's no duplication.

2. Meal Plans

Number of types: 4 meal plans offered.

Most popular: Meal Plan 1 with 27,835 bookings (~77% of total).

Observation: The majority of guests prefer Meal Plan 1, which suggests operational focus can be on this plan for food inventory and service planning.

3. Room Types

Number of types: 7 room types.

Most popular: Room_Type 1 with 28,130 bookings (~77% of total).

Observation: One room type dominates, indicating a concentration of demand. Other room types may need targeted marketing or promotions to increase occupancy.

4. Market Segment

Number of segments: 5 segments.

Most bookings via: Online (23,214 bookings, ~64% of total).

Observation: Online channels drive most bookings, highlighting the importance of maintaining strong digital presence. Offline channels contribute less, potentially an area for strategic growth.

5. Booking Status (Target Variable)

Not Canceled: 24,390 bookings (~67%).

Canceled: 12,885 bookings (~33%).

Observation: About 1 in 3 bookings get canceled, indicating a significant cancellation rate. Understanding the features that influence cancellations (lead time, room type, market segment, etc.) will be critical for revenue and operational planning.

Checking for duplicate values

	0
Booking_ID	0
no_of_adults	0
no_of_children	0
no_of_weekend_nights	0
no_of_week_nights	0
type_of_meal_plan	0
required_car_parking_space	0
room_type_reserved	0
lead_time	0
arrival_year	0
arrival_month	0
arrival_date	0
market_segment_type	0
repeated_guest	0
no_of_previous_cancellations	0
no_of_previous_bookings_not_canceled	0
avg_price_per_room	0
no_of_special_requests	0
booking_status	0

Figure 5: Duplicate values

There are no duplicate values in the data

Checking for missing values

	0
Booking_ID	0
no_of_adults	0
no_of_children	0
no_of_weekend_nights	0
no_of_week_nights	0
type_of_meal_plan	0
required_car_parking_space	0
room_type_reserved	0
lead_time	0
arrival_year	0
arrival_month	0
arrival_date	0
market_segment_type	0
repeated_guest	0
no_of_previous_cancellations	0
no_of_previous_bookings_not_canceled	0
avg_price_per_room	0
no_of_special_requests	0
booking_status	0

dtype: int64

Figure 6: Checking for missing values

There are no missing values in the dataset.

Exploratory Data Analysis (EDA) Summary

Univariate Analysis

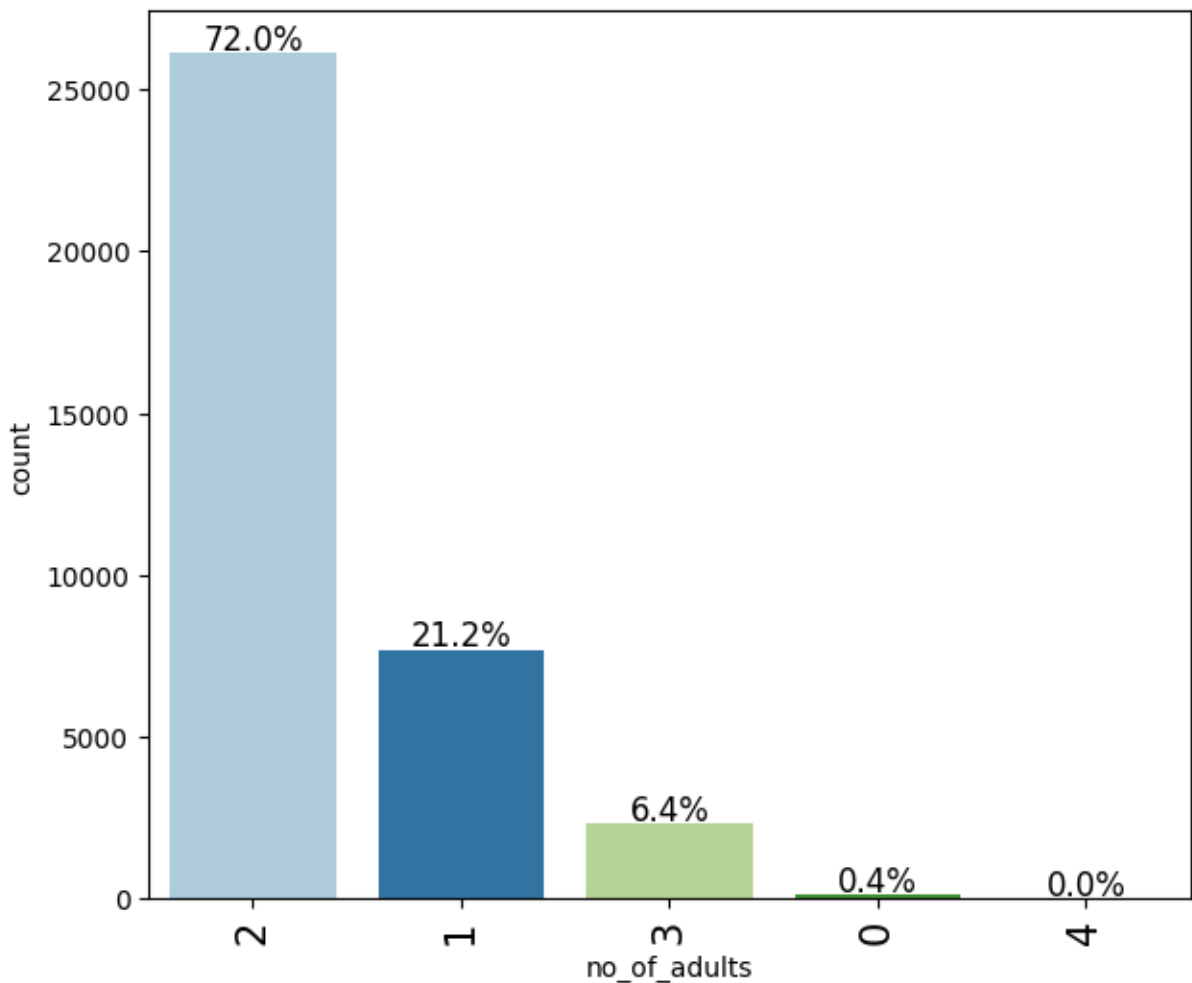


Figure 7: No. of Adults

- 2 adults: 72% of bookings → Most bookings are for couples or pairs, representing the core guest segment.
- 1 adult: 21.2% → Significant minority of solo travelers.
- 3 adults: 6.4% → Less common; could represent small family or group travel.
- No adults: 0.4% → Extremely rare; may indicate data entry anomalies or special cases.
- Observation: The hotel primarily caters to couples and solo travelers, so room configurations, packages, and promotions can be designed to appeal to these groups.

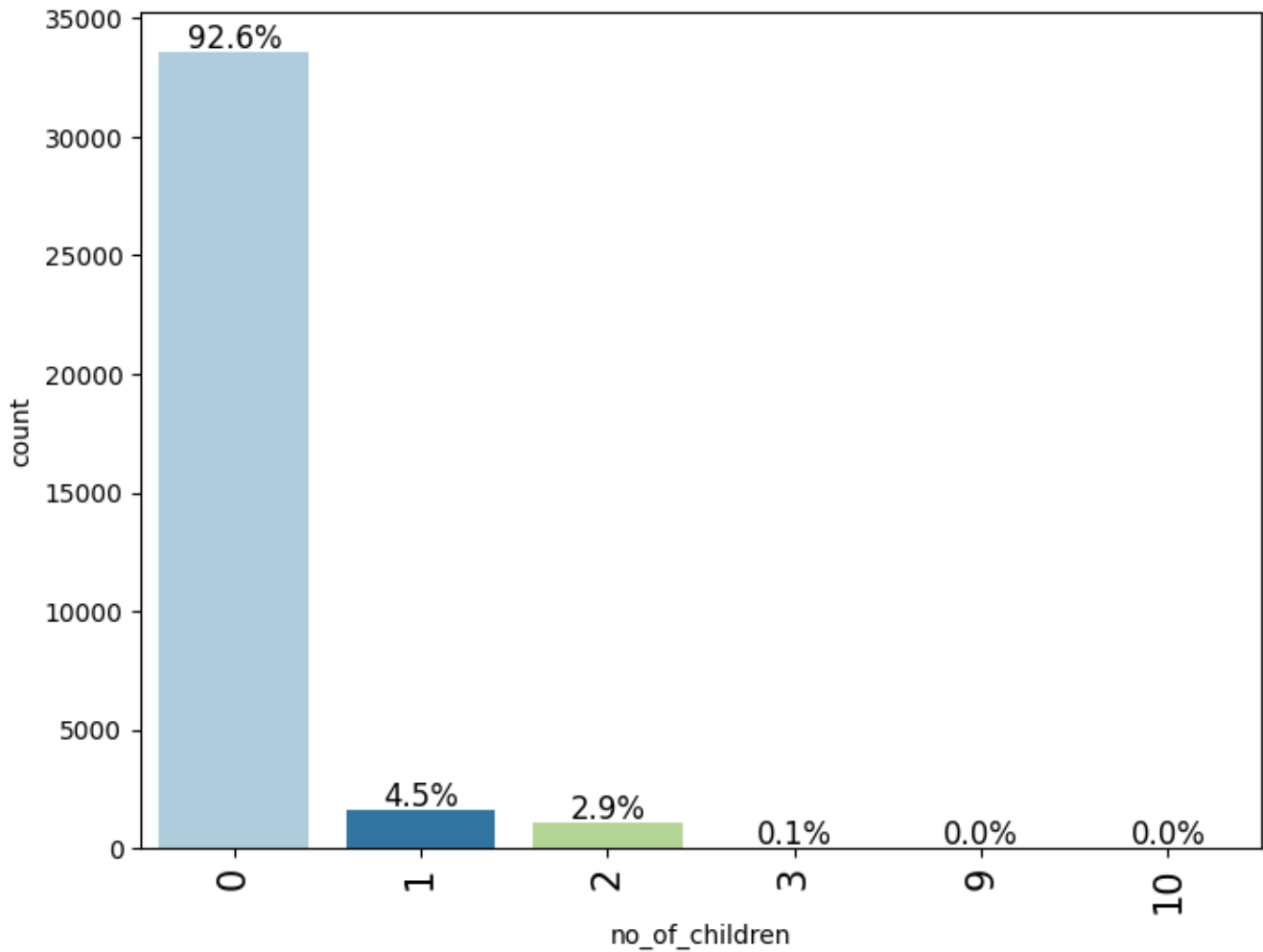


Figure 8: No. of Children

- No children: 92.6% → Vast majority of bookings are adult-only, indicating that the hotel primarily serves couples and adult travelers.
- 1 child: 4.5% → Small fraction of families with one child.
- 2 children: 2.9% → Few family bookings with two children.

Observation: Family bookings are a minor segment. Marketing, room configuration, and services can focus on adult travelers, while targeted family packages can be designed to attract this smaller segment.

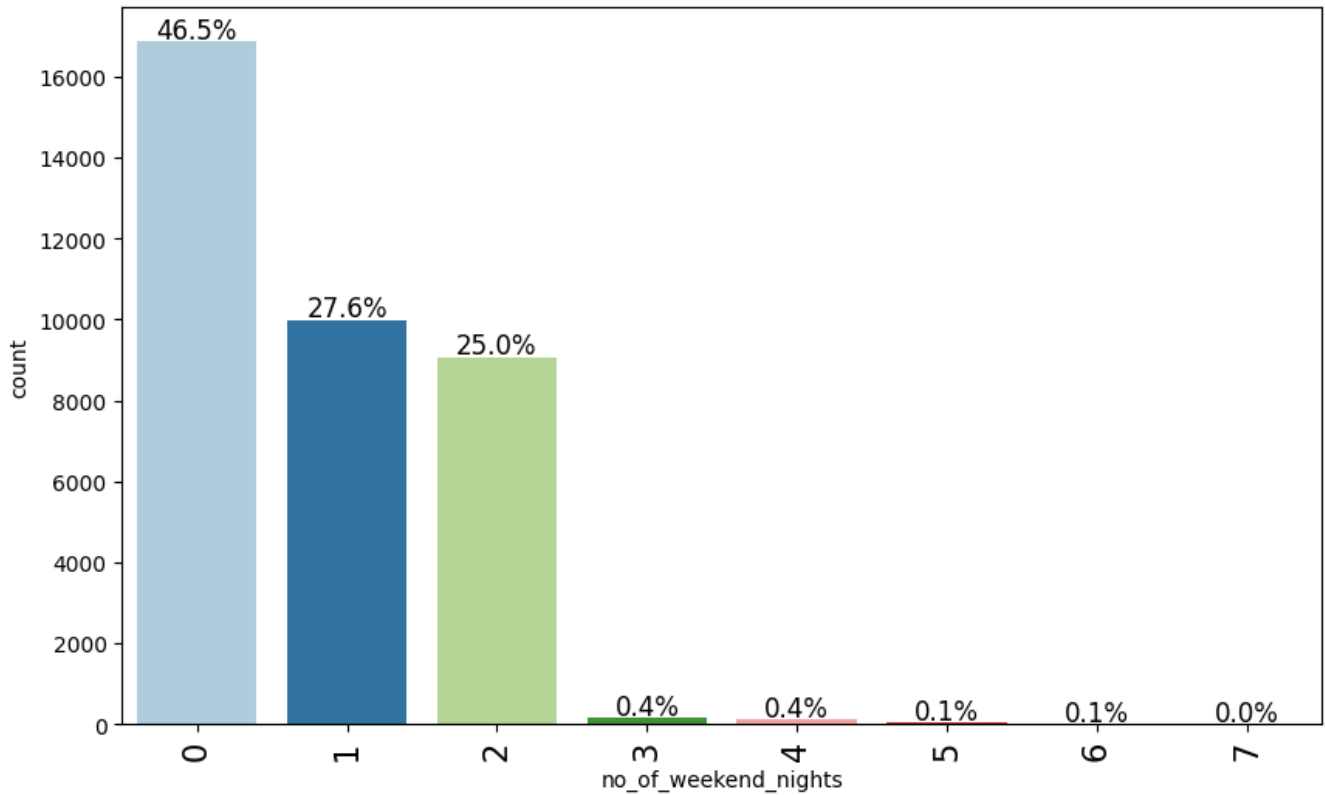


Figure 9: No. of Weekend Nights

- 0 weekend nights: 46.5% → Almost half of bookings are for weekday-only stays, indicating a strong business or midweek traveler segment.
- 1 weekend night: 27.6% → Over a quarter include short weekend stays.
- 2 weekend nights: 25% → A quarter of bookings span the full weekend.

Observation: While weekday stays dominate, a significant portion of bookings occur over weekends. This suggests the hotel should balance staffing and pricing strategies between weekdays and weekends to optimize occupancy and revenue.

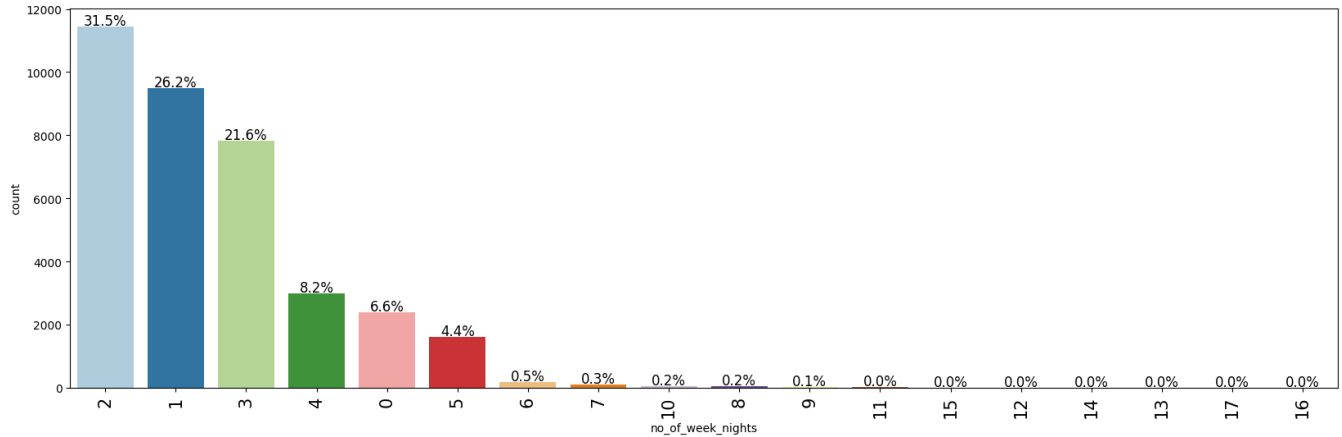


Figure 10: No. of Week Nights

- 2 week nights: 31.5% → The most common stay length, indicating short midweek trips are typical.
- 1 week night: 26.2% → A significant portion of very short weekday stays, likely business travelers.
- 3 week nights: 21.6% → Considerable number of slightly longer stays.

Observation: The majority of bookings are for 1–3 week nights, reinforcing that the hotel primarily accommodates short-duration stays. Combined with weekend data, this suggests a mix of midweek business travelers and weekend leisure guests, which can guide staffing, room allocation, and promotional strategies.

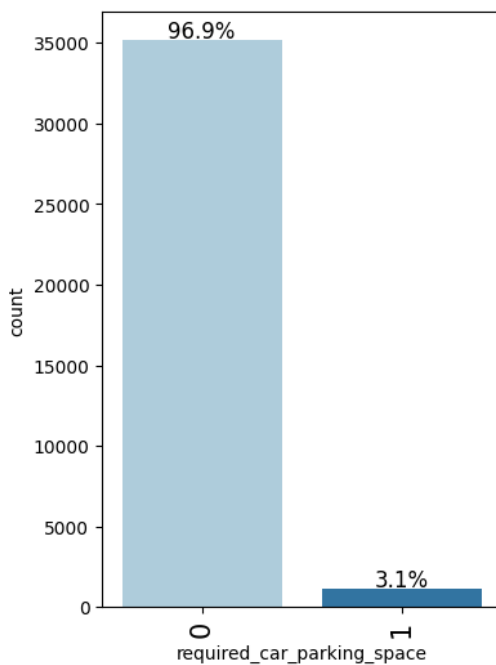


Figure 11: Required Car Parking Space

- Bookings requiring parking: 3.1% (1,124 reservations) → Very few guests need parking.

Observation: Parking is a minor requirement for most guests. Operational focus can remain on room and meal services, with parking management scaled to a small subset of bookings..

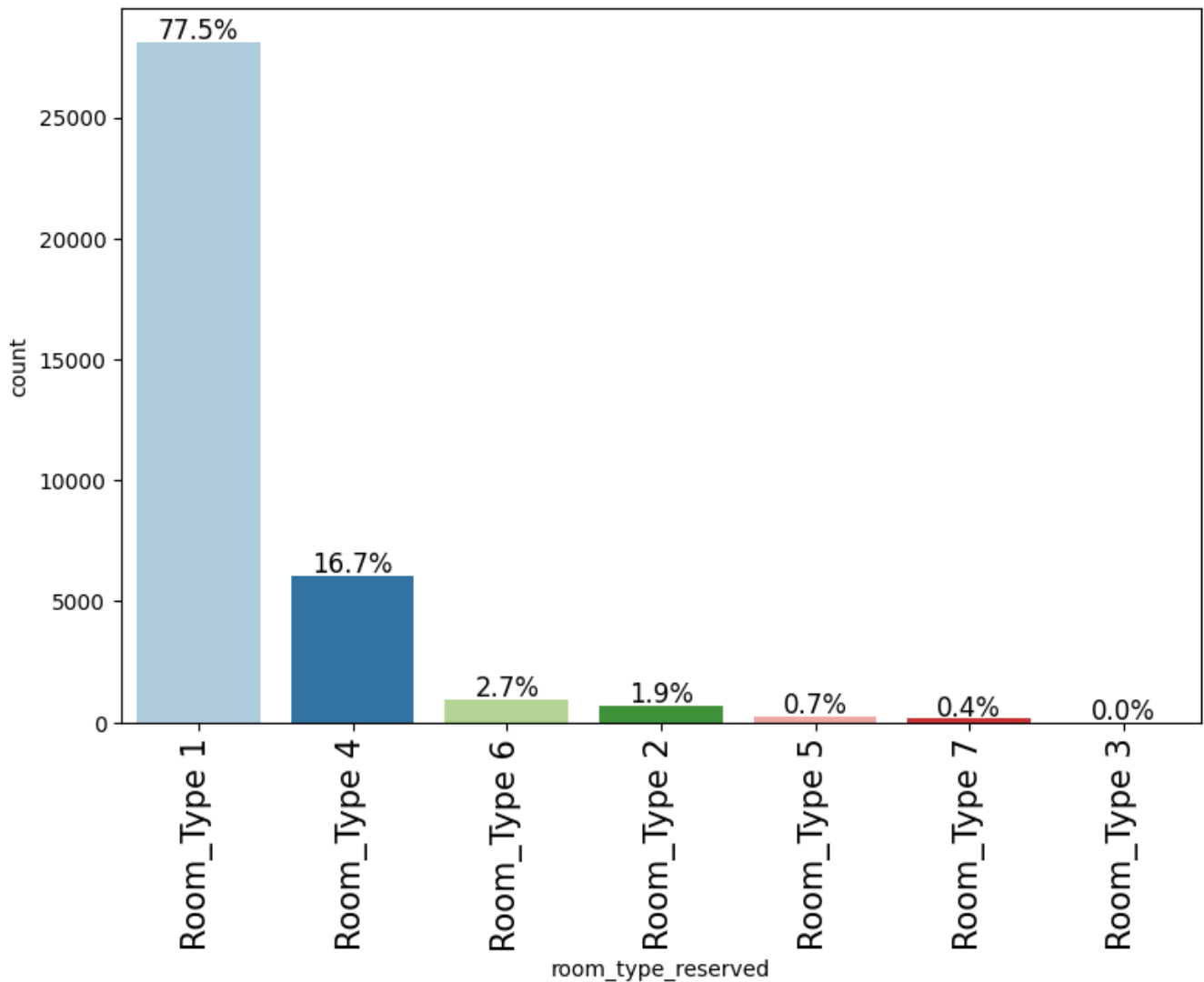


Figure 12: Room Type Reserved

Room Type 1: 77.5% → The overwhelming majority of guests prefer this room type.

Observation: Demand is highly concentrated on Room Type 1.

This indicates opportunities to:

- Optimize allocation and pricing for this room type.
- Promote alternative room types to balance occupancy and increase revenue from less-booked categories.

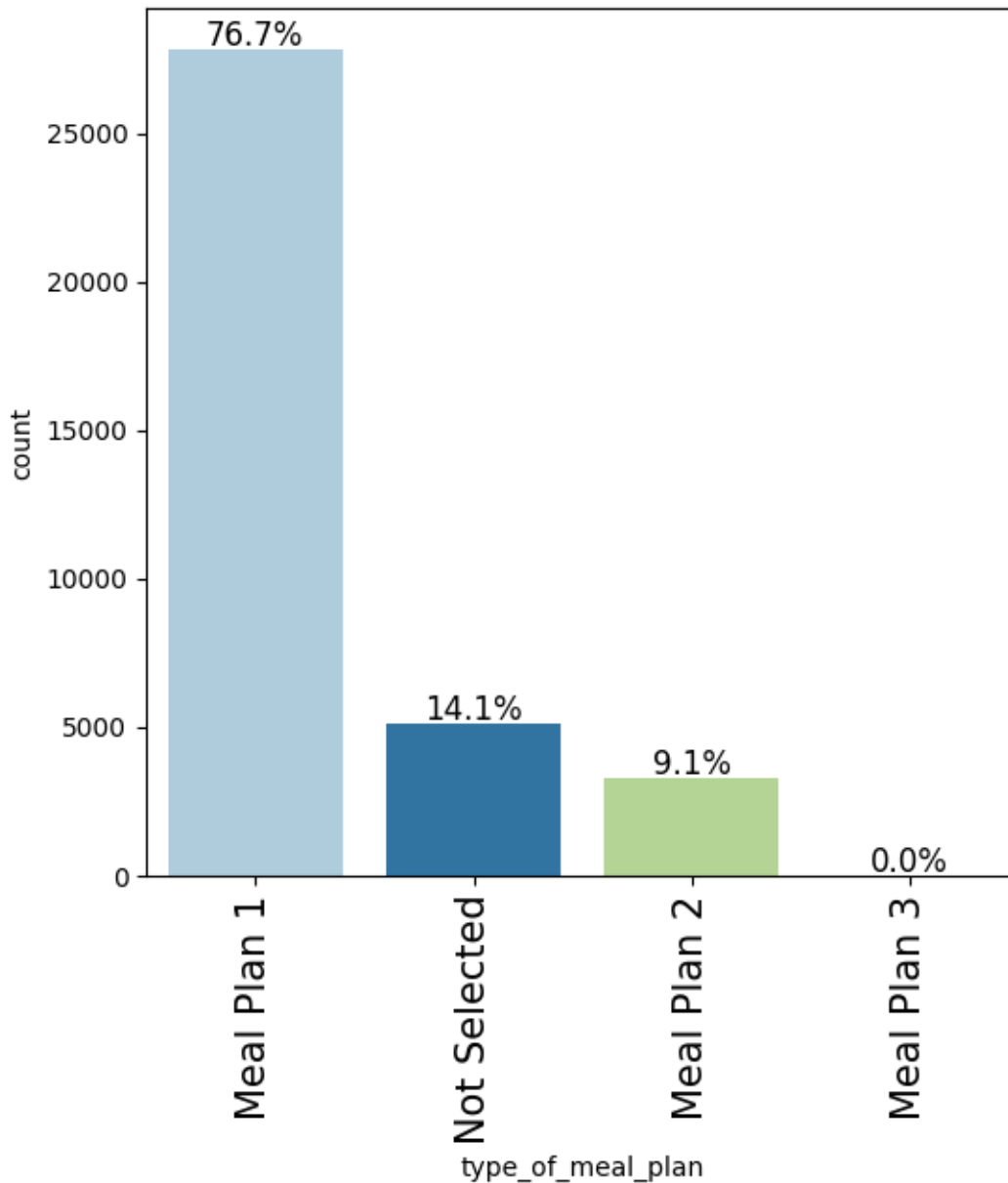


Figure 13: Type of Meal Plan

- Meal Plan 1 (Breakfast only): 76.7% → Clearly the most popular option, indicating that most guests prefer minimal meal inclusion, likely to have flexibility or dine outside.
- Meal Plan 2 (Breakfast + 1 other meal): 14.1% → A smaller segment chooses a slightly more inclusive plan, suggesting some demand for convenience or half-board options.

Observation: The hotel can prioritize operations around Meal Plan 1, ensuring efficiency and cost management. Marketing strategies or packages could be designed to upsell Meal Plan 2 or other plans to increase revenue from guests willing to pay for additional meals.

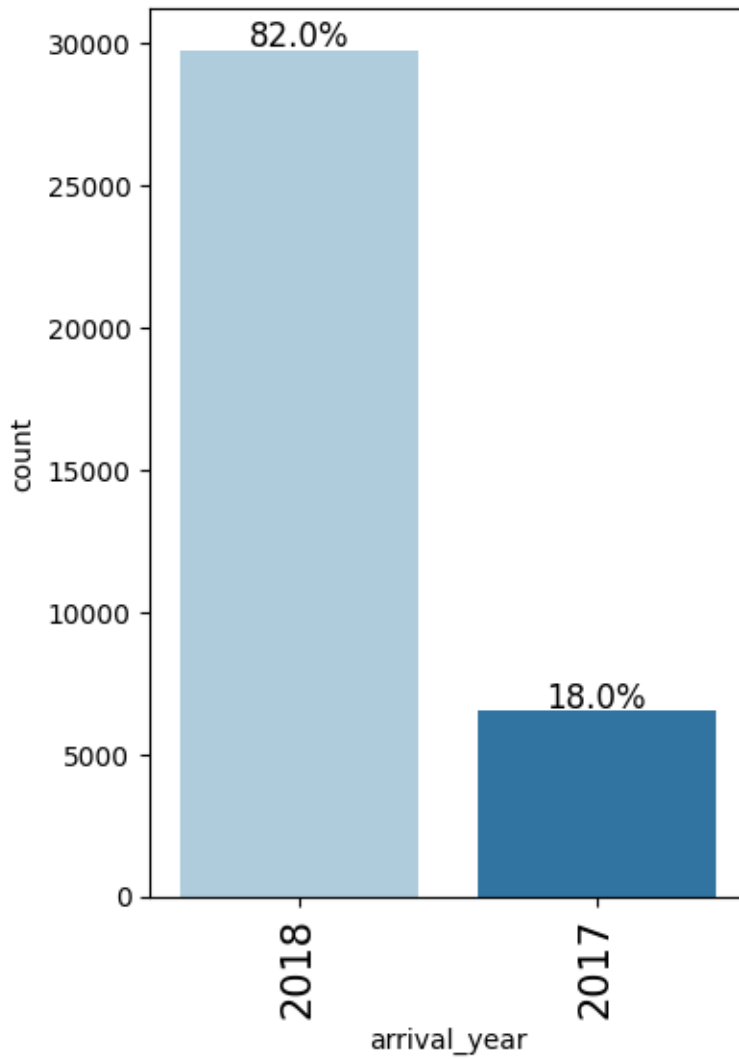


Figure 14: Arrival Year

2018 bookings: ~82% of total bookings.

Observation: The dataset is heavily concentrated in 2018, indicating that the majority of the analysis reflects recent trends. Any strategic decisions or operational insights drawn will primarily apply to the 2018 booking patterns.

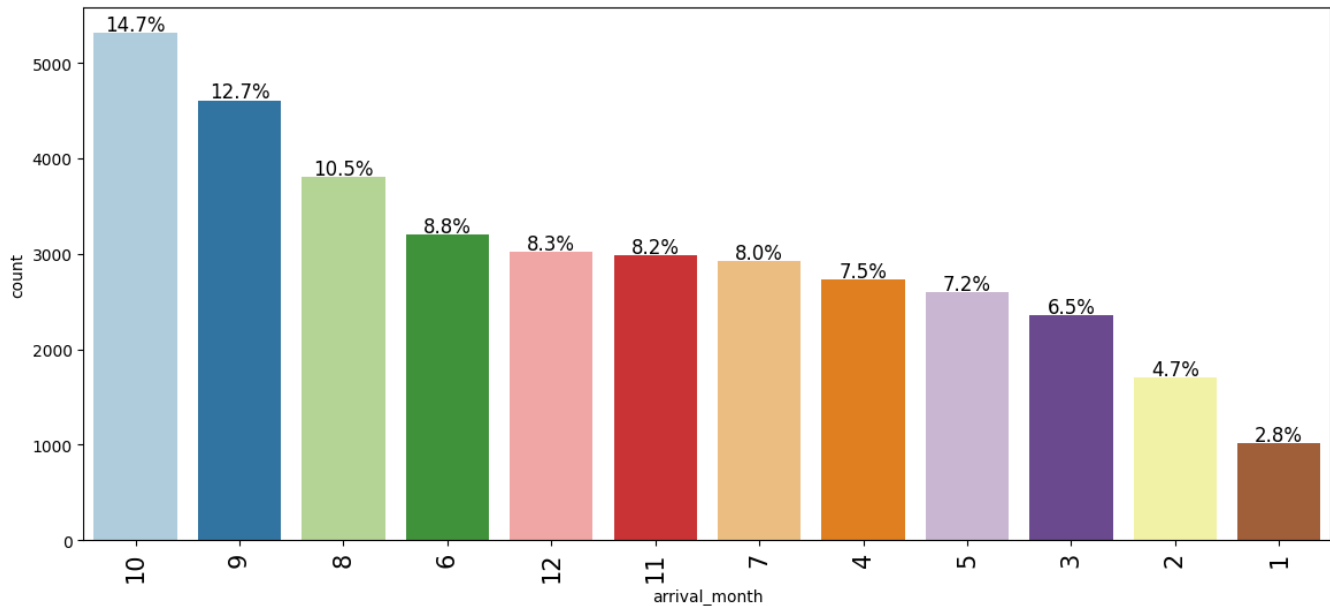


Figure 15: Arrival Month

Seasonality & Arrival Patterns

Peak booking months:

August: 10.5% of bookings → Start of high season.

October: 14.7% of bookings → Peak period, likely driven by popular travel season or events.

Post-peak trends:

November: 8.2% → Still relatively high.

December–February: Significant decline, with January lowest at 2.8%, possibly due to unfavorable winter weather.

Observation:

There is a clear seasonal pattern in bookings.

The hotel can optimize staffing, promotions, and pricing strategies to capitalize on the high-demand period (August–October) and manage lower occupancy months efficiently.

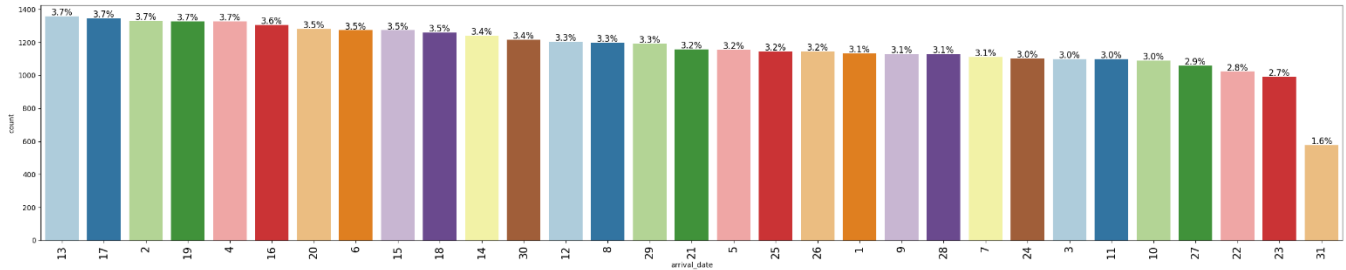


Figure 16: Arrival Date

Arrival Date Analysis

Overall distribution: Fairly uniform across days of the month, indicating no strong preference for specific dates.

Notable exceptions:

31st of the month: Only 1.6% of arrivals → Limited by the fact that only 7 months have 31 days.

Most popular dates: 13th (1,358 arrivals), followed by 2nd, 4th, 17th, and 19th → Slight preference for mid-month bookings.

Observation:

The hotel can expect relatively consistent daily arrivals, simplifying day-to-day operational planning.

Minor peaks on specific dates may allow for targeted staffing adjustments or promotions to manage workload efficiently.

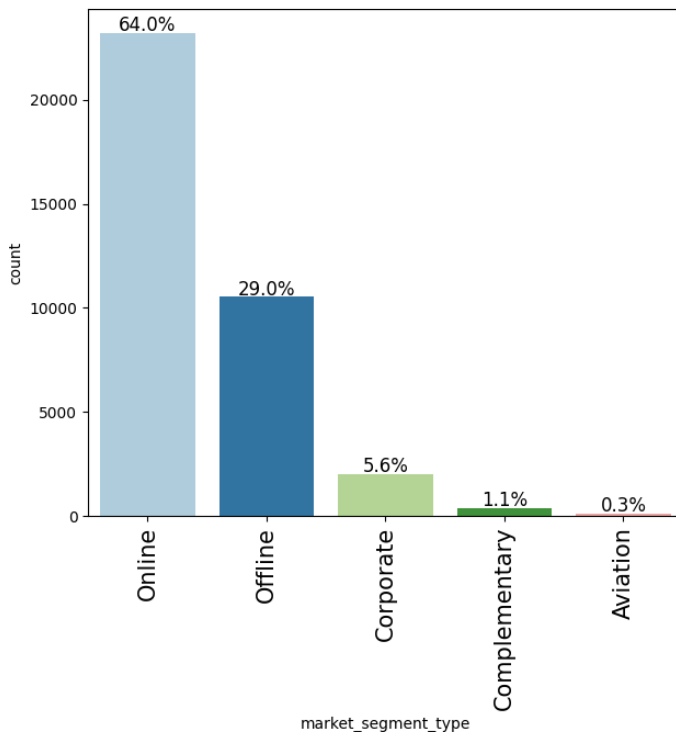


Figure 17: Market Segment Type

Market Segment Analysis

Online bookings: 63.99% → Majority of guests book through online channels, highlighting the importance of digital presence and online marketing.

Offline bookings: 29.02% → Significant proportion still comes through offline channels (telephone or walk-ins),

suggesting the hotel should maintain efficient offline booking processes.

Corporate bookings: 5.56% → Small but notable segment, likely for business travel, which could be further targeted with tailored corporate packages.

Complementary bookings: 1.08% → Very low, may represent special promotions or partner-driven bookings.

Aviation bookings: 0.34% → Minimal contribution, possibly through airline partnerships.

Observation:

The hotel relies heavily on online channels, but offline and corporate segments still represent important niches. Opportunities exist to grow corporate and complementary bookings through targeted marketing and partnerships.

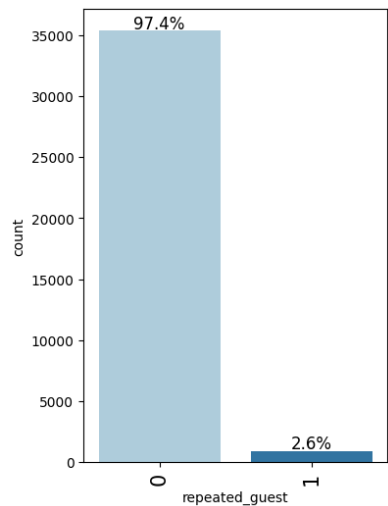


Figure 18: Repeated Guest

Guest Loyalty Analysis

One-time guests: 97.44% (35,345 bookings) → The vast majority of guests are first-time or occasional visitors, indicating limited repeat business.

Repeated guests: 2.56% (930 bookings) → A small segment demonstrates loyalty, choosing to book more than once, which could reflect satisfaction with the hotel or booking platform.

Observation:

Guest loyalty is currently low, presenting an opportunity to implement loyalty programs, personalized offers, or retention strategies to encourage repeat bookings.

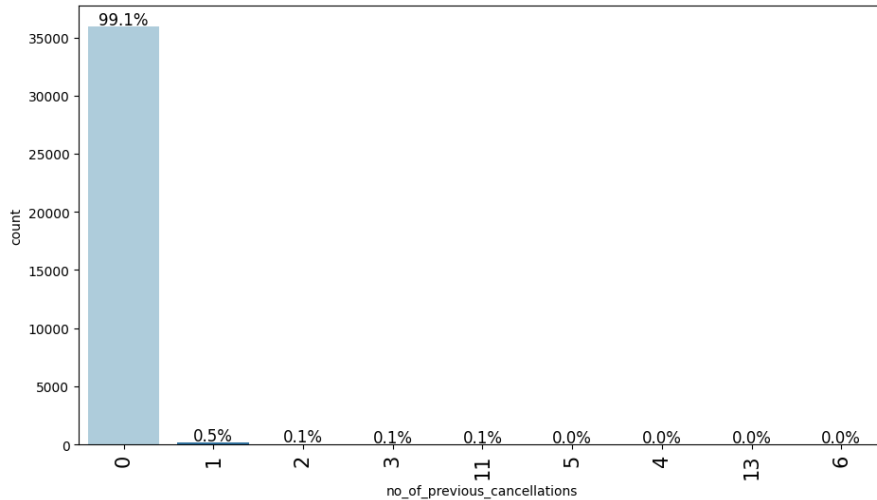


Figure 19: No. of Previous Cancellation

Booking Cancellation History

No previous cancellations: 95.2% → Most guests have a clean booking history, suggesting high reliability.

1 previous cancellation: 0.5% (198 guests) → Small fraction with prior cancellation experience.

Multiple cancellations: Very few guests (~140) have canceled more than twice, indicating a very small high-risk group.

Observation:

The vast majority of guests are reliable, with low historical cancellation rates.

Management can focus cancellation mitigation strategies on the small subset with a history of multiple cancellations to protect revenue.

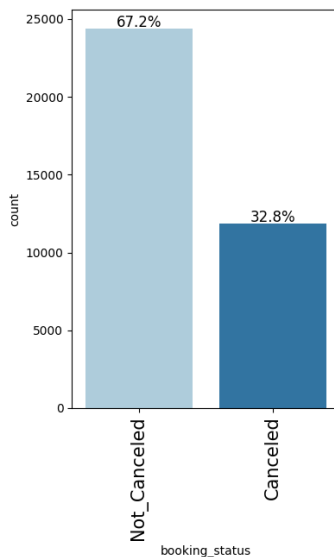


Figure 20: Booking Status

Booking Status

Not canceled: 67.2% → Majority of bookings are honored, indicating fairly stable occupancy.

Canceled: 32.8% → Nearly 1 in 3 bookings gets canceled, representing a significant revenue risk.

Observation:

Cancellation rate is substantial, so the hotel should consider strategies such as:

Flexible deposits or partial prepayments to reduce cancellations.

Targeted communication with high-risk bookings (long lead times, multiple previous cancellations).

Dynamic pricing and overbooking strategies to mitigate revenue loss from cancellations.

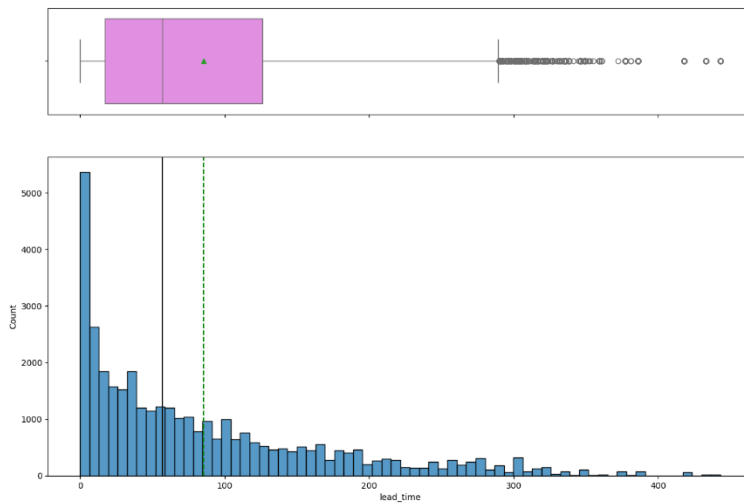


Figure 21: Lead Time

- Average lead time: 85 days → On average, guests book about three months in advance.
- Distribution: Right-skewed with outliers → Some guests book well in advance, up to 443 days.
- 75th percentile: 126 days → Most guests (75%) book within four months of their arrival.

Observation:

- The majority of bookings are made within a short to medium-term window, allowing the hotel to forecast occupancy and revenue reasonably accurately.
- Long lead bookings are outliers, which may carry a higher risk of cancellations and should be monitored for revenue management strategies.

Bivariate Analysis



Figure 22: Heat Map

Correlation Insights

- Adults and children vs. room price:
- no_of_adults and avg_price_per_room → moderate positive correlation (~0.30).
- no_of_children and avg_price_per_room → moderate positive correlation (~0.35).

Interpretation: Larger groups, whether adults or children, tend to book higher-priced rooms, likely due to needing bigger or more premium accommodations.

- Stay duration and lead time vs. room price:
- no_of_weekend_nights and no_of_week_nights positively correlate with avg_price_per_room and lead_time.
- lead_time and avg_price_per_room → moderate positive correlation (~0.44).
- Interpretation: Longer stays and bookings made further in advance are associated with higher room prices, which may reflect premium pricing for long-term or early reservations.

Parking requirements:

- Slight negative correlation with avg_price_per_room.
- Interpretation: Guests requiring parking may slightly prefer lower-priced rooms.

Repeated guests:

- Negative correlation with no_of_adults, no_of_children, and no_of_special_requests.
- Interpretation: Loyal guests tend to book smaller parties and request fewer special services, indicating familiarity and simplicity in their booking behavior.

Booking status:

- Weak positive correlation with lead_time and avg_price_per_room.
- Interpretation: Bookings with longer lead times and higher prices have a slightly higher risk of cancellation, though correlation is weak.

Business Insight:

- Larger and longer bookings generate higher revenue, but long lead-time bookings may need cancellation mitigation strategies.
- Understanding guest composition and booking behavior can help tailor pricing, promotions, and operational planning.

Note: Correlation indicates association, not causation, and should be interpreted as indicative trends rather than direct effects.

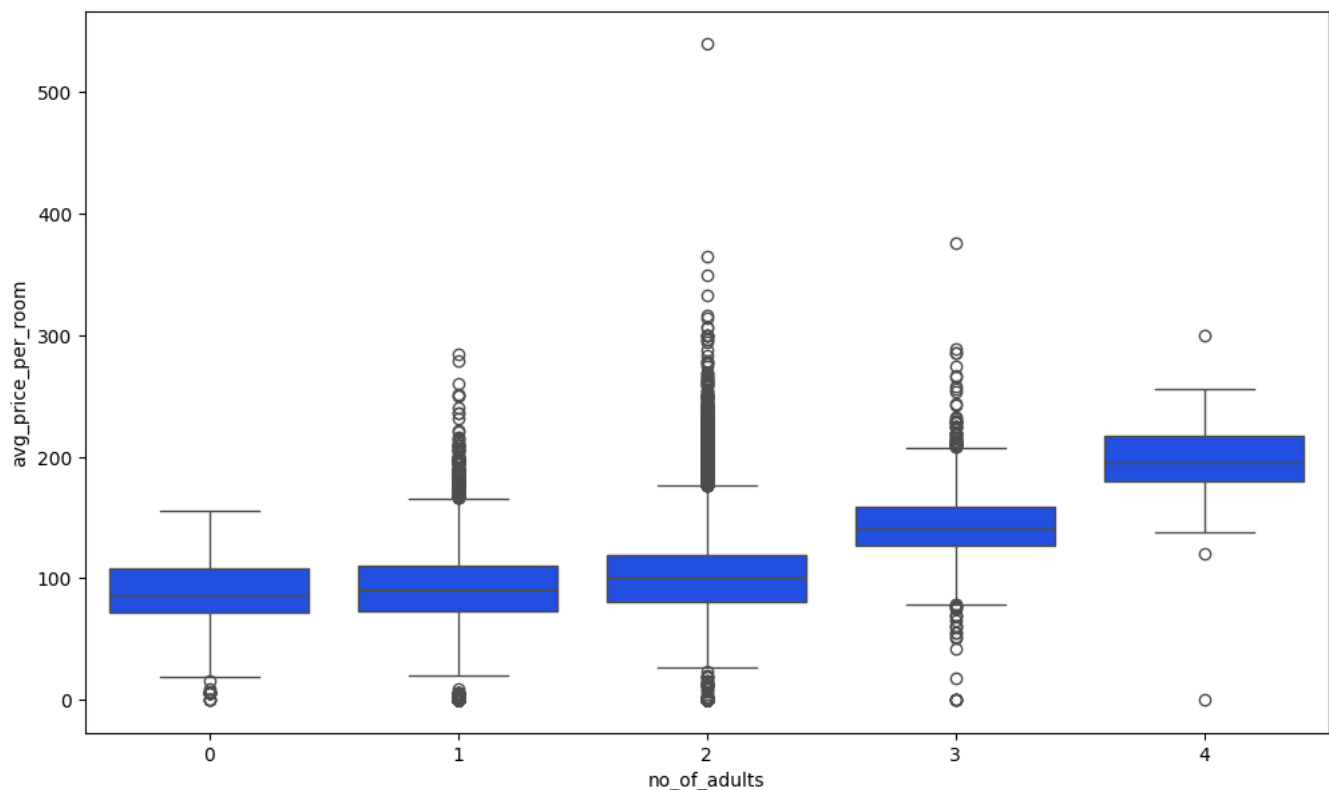


Figure 23: No. of Adults vs Avg. Price per Room

Observation: There is a clear correlation between number of adults and average room price. As the number of adults increases, the room price tends to increase, likely due to the need for larger or premium rooms.

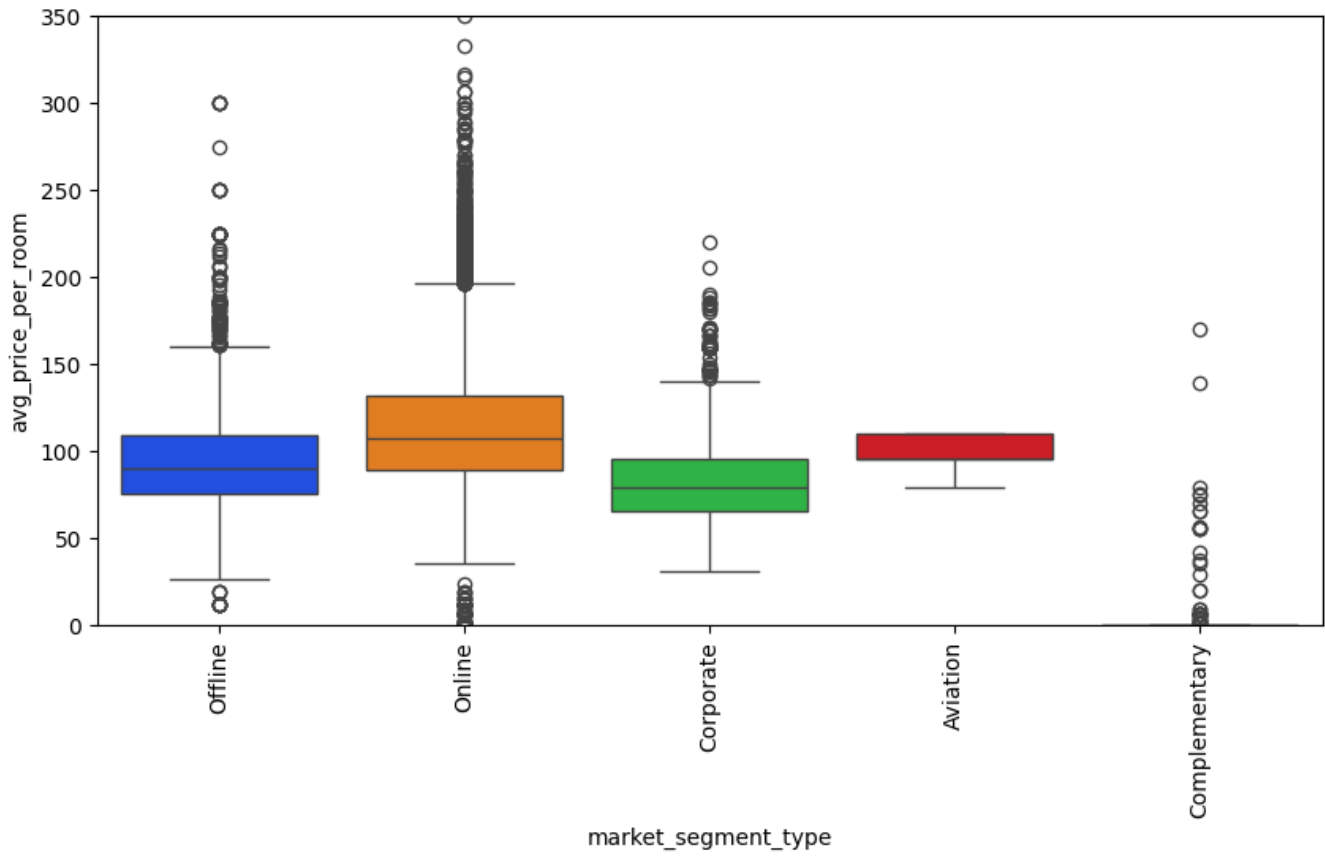


Figure 24: Avg. Price per Room Vs Market Segment Type

- Online channels attract guests willing to pay higher average prices, providing an opportunity to optimize pricing and promotional strategies for this segment.
- Wide price variation suggests that dynamic pricing or targeted upselling could further maximize revenue from online bookings.

booking_status	Canceled	Not_Canceled	All
market_segment_type			
All	11885	24390	36275
Online	8475	14739	23214
Offline	3153	7375	10528
Corporate	220	1797	2017
Aviation	37	88	125
Complementary	0	391	391

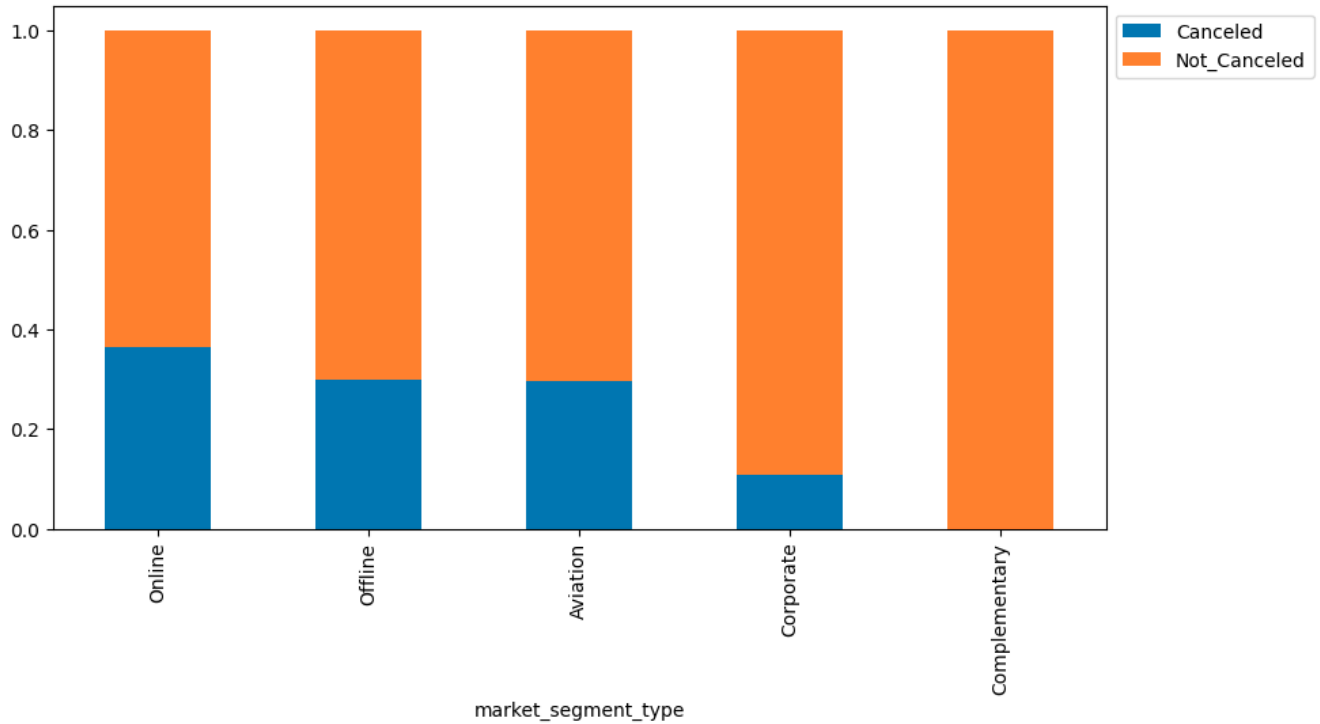


Figure 25: Market Segment Type vs Booking Status

Cancellations by Market Segment

- Online: 23,214 bookings → 14,739 not canceled, 8,475 canceled.
- Offline: 10,528 bookings → 7,375 not canceled, 3,153 canceled.
- Corporate: 2,017 bookings → 1,797 not canceled, 220 canceled.
- Aviation: 125 bookings → 88 not canceled, 37 canceled.
- Complementary: 391 bookings → all not canceled.

Observation:

Online and offline segments show the highest number of cancellations.

Complementary bookings are fully reliable, while corporate and aviation segments have relatively low cancellation rates.

no_of_special_requests	0	1	2	3	4	5	All
market_segment_type							
All	19777	11373	4364	675	78	8	36275
Online	9251	9151	4113	628	64	7	23214
Offline	8621	1710	173	20	3	1	10528
Aviation	125	0	0	0	0	0	125
Complementary	167	144	50	19	11	0	391
Corporate	1613	368	28	8	0	0	2017

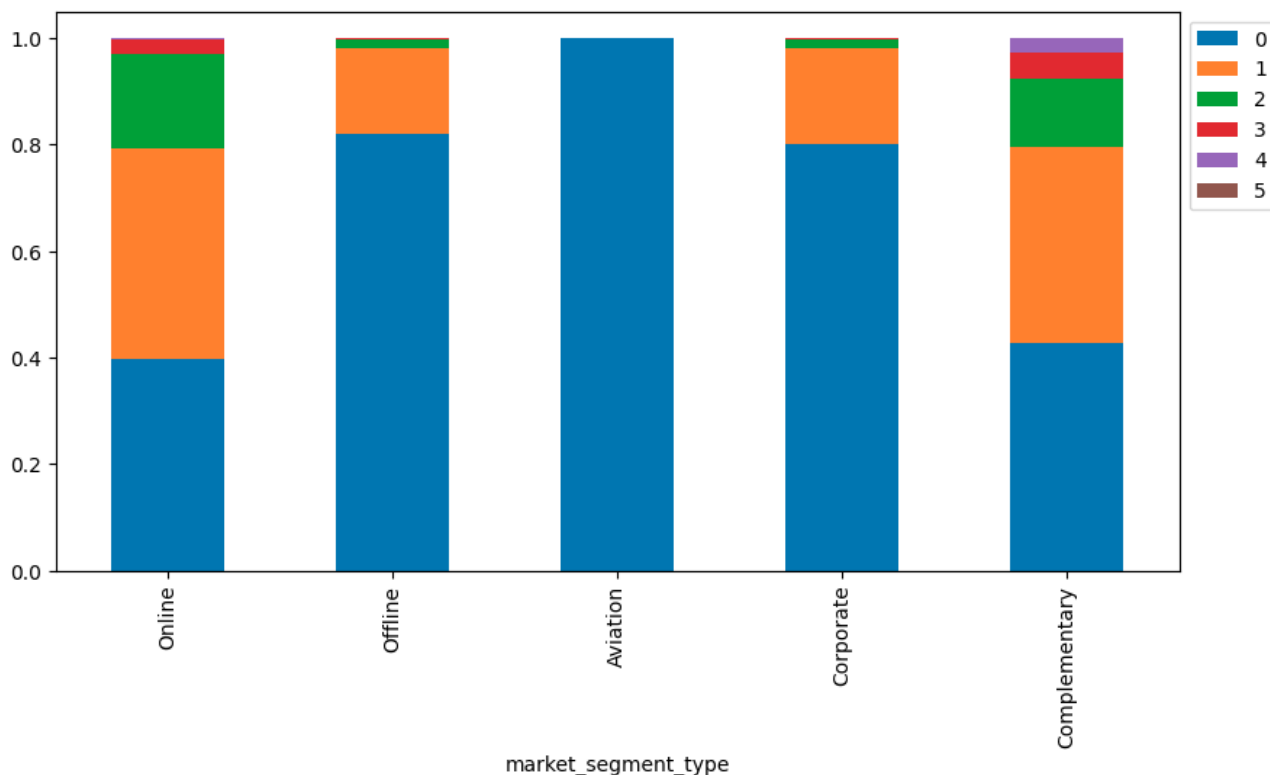


Figure 26: Market Segment Type vs No. of Special Request

Special Requests by Market Segment

- Overall: Most bookings (19,777) have no special requests, while very few bookings have 4–5 requests.
- Online: High volume (9,251) with no requests, but also substantial bookings with 1–2 requests, showing some guests personalize their stay.
- Offline: Majority (8,621) have no requests; very few have multiple requests.
- Corporate & Complementary: Small numbers, mostly 0–1 special requests.

Observation:

Most guests do not make special requests, but online bookings show more variability, indicating potential for upselling services or customizing experiences.

Figure 27: No. of Special Requests vs Avg. Price per Room

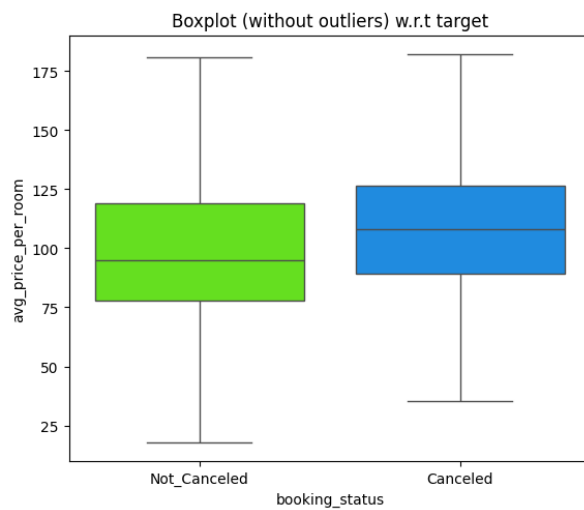
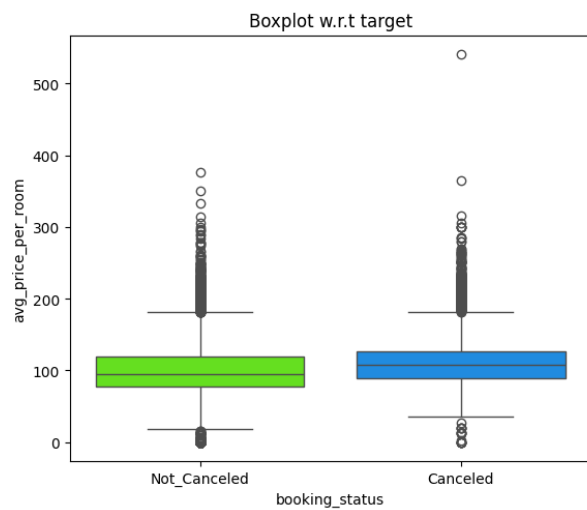
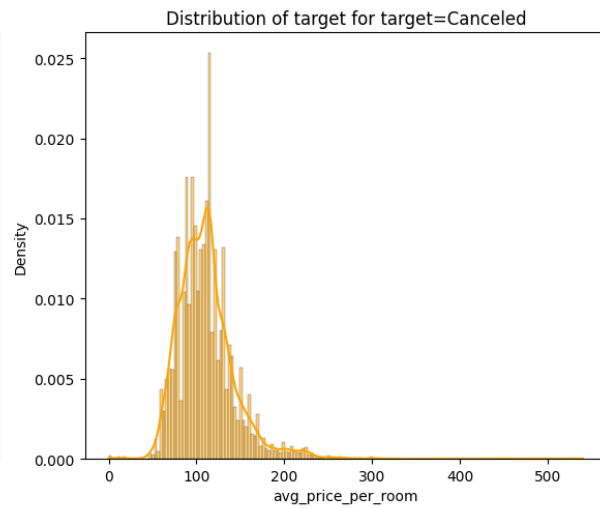
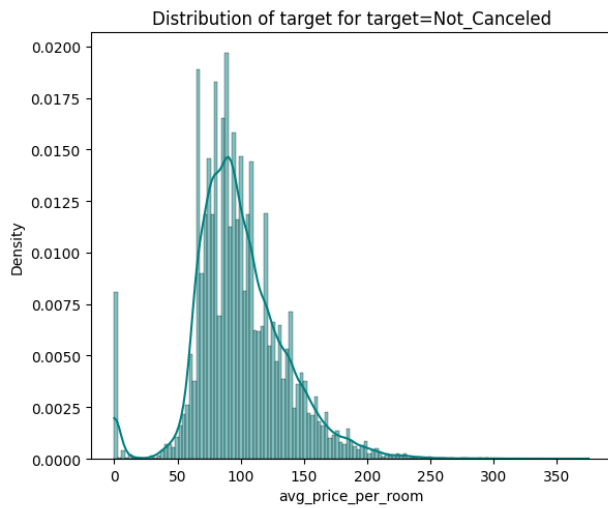
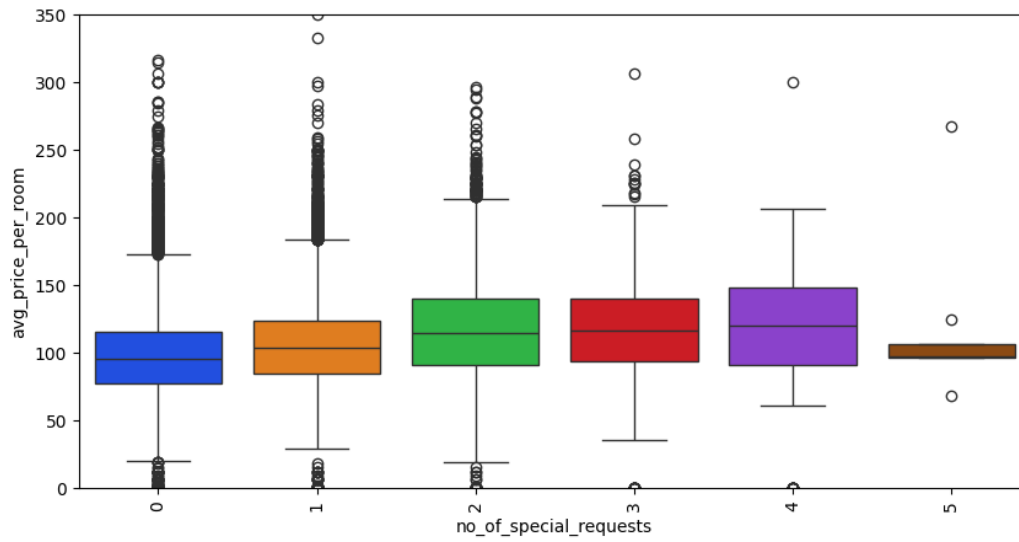


Figure 28: Distribution and Boxplot graph_1

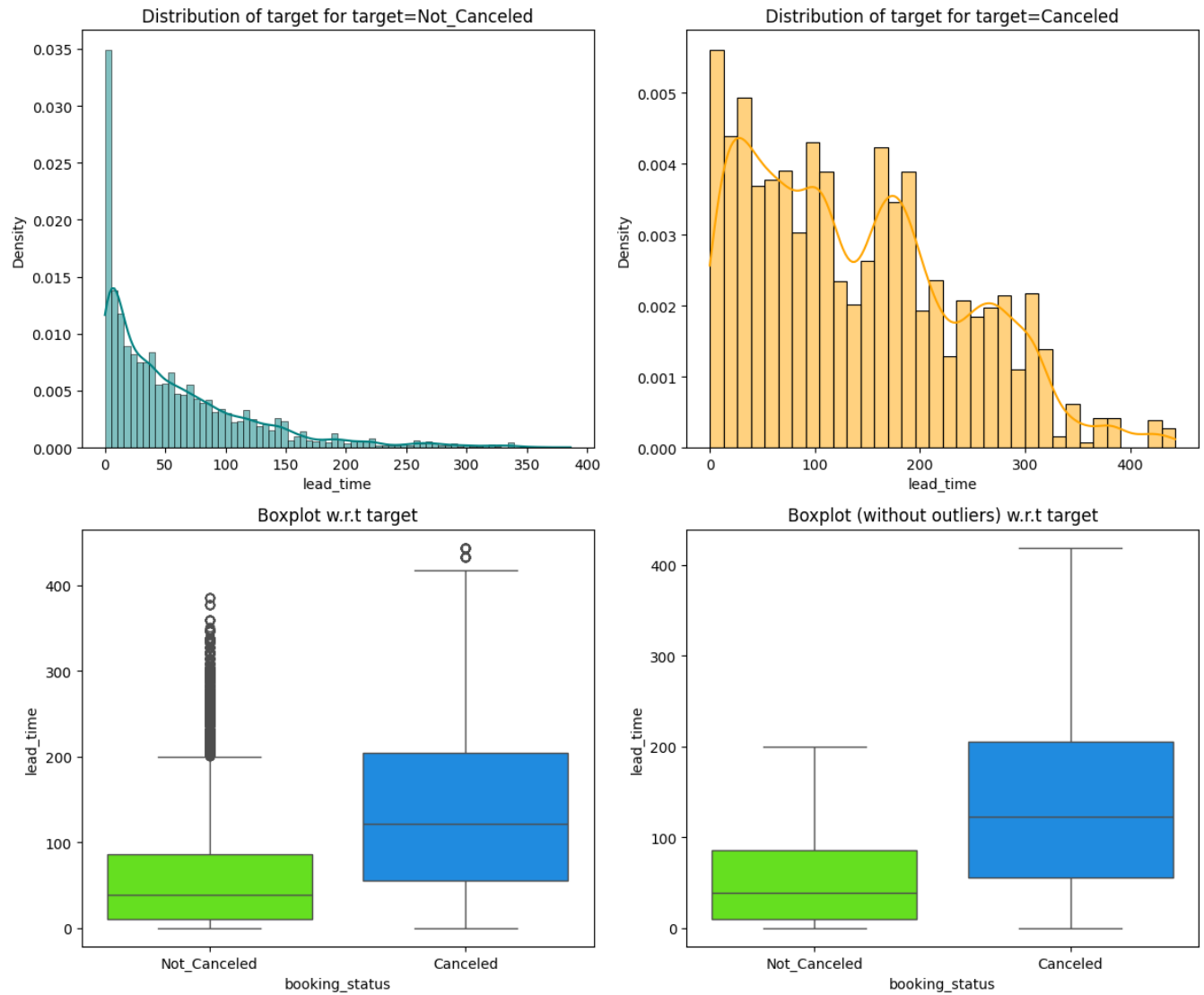


Figure 29: Distribution and Boxplot graph_2

booking_status	Canceled	Not_Canceled	All
no_of_family_members			
All	9985	18456	28441
2	8213	15506	23719
3	1368	2425	3793
4	398	514	912
5	5	10	15
11	1	0	1
12	0	1	1

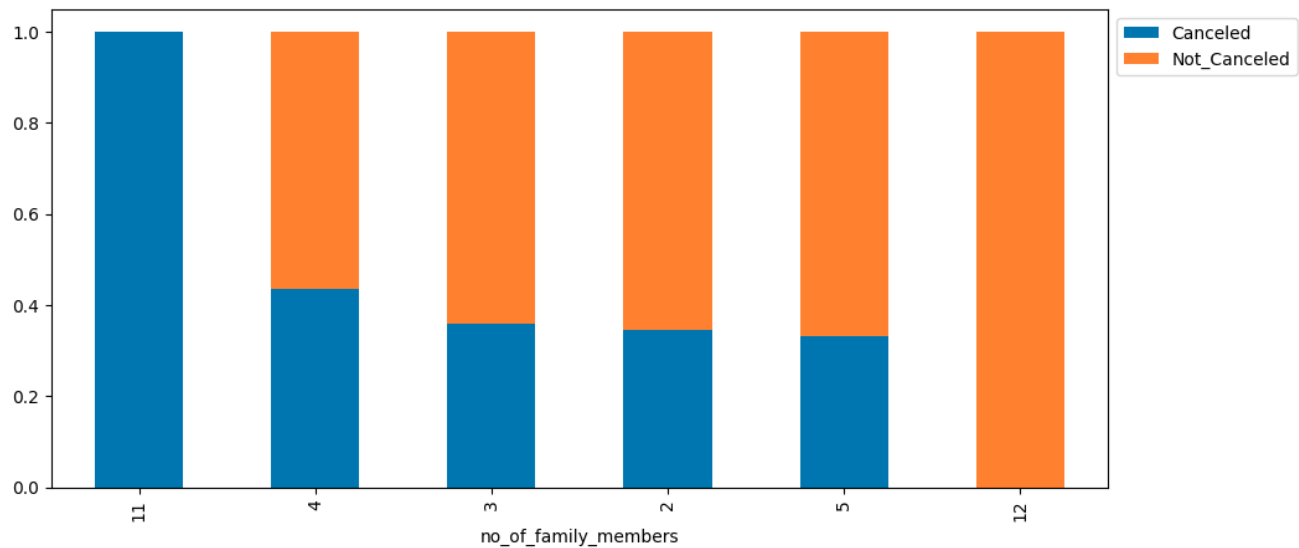


Figure 30: No. of Family members vs Booking Status

Cancellations by Number of Family Members

- Overall: 9,985 bookings were canceled, 18,456 not canceled.
- Most common family sizes:
- 2 members → 8,213 canceled, 15,506 not canceled.
- 3 members → 1,368 canceled, 2,425 not canceled.
- Larger groups (4–5 members) show very few bookings, with minimal cancellations.

Observation:

Smaller families (2–3 members) make up the majority of bookings and cancellations. Very large groups are rare.

booking_status	Canceled	Not_Canceled	All
no_of_special_requests			
All	11885	24390	36275
0	8545	11232	19777
1	2703	8670	11373
2	637	3727	4364
3	0	675	675
4	0	78	78
5	0	8	8

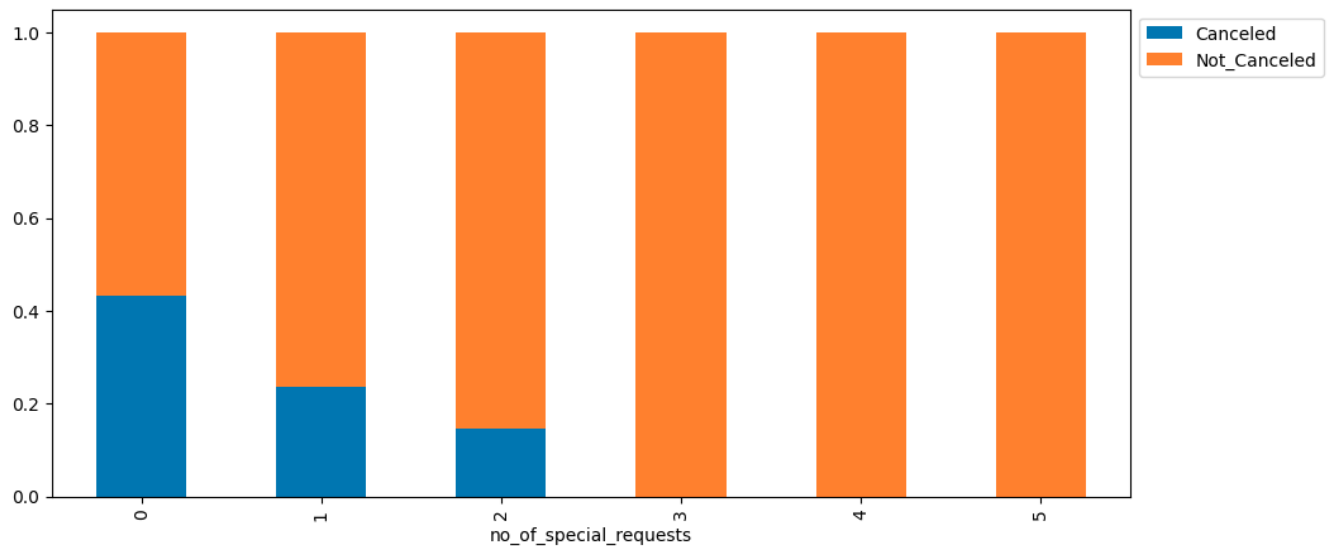


Figure 31: No. of Special Requests vs Booking Status

Cancellations vs. Special Requests

- Overall: 11,885 bookings canceled, 24,390 not canceled.
- No special requests (0): 8,545 canceled, 11,232 not canceled → Most cancellations occur among guests with no special requests.
- 1–2 special requests: Moderate cancellations (2,703 with 1 request, 637 with 2 requests).
- 3–5 special requests: Almost no cancellations → Guests making multiple special requests tend to honor their bookings.

Observation:

Guests with more special requests are less likely to cancel, suggesting that personalized bookings indicate higher commitment.

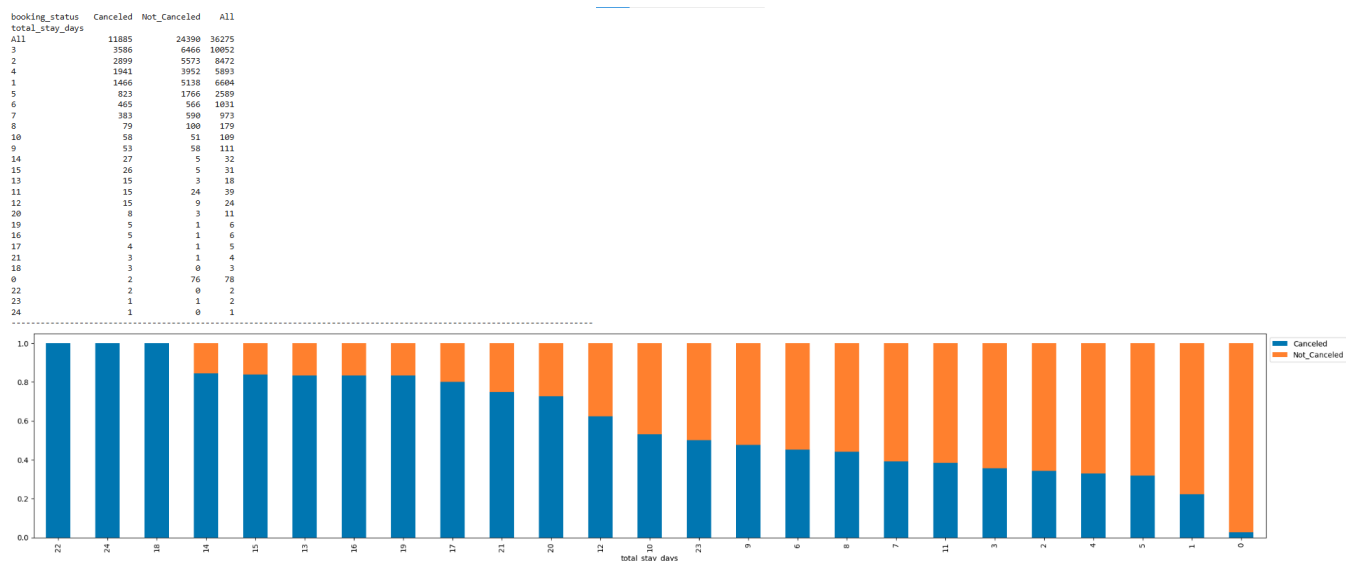


Figure 32: Total Stay Days vs Booking Status

Cancellations vs. Total Stay Duration

Overall: 11,885 bookings canceled, 24,390 not canceled.

Most common stays:

- 3 days → 3,586 canceled, 6,466 not canceled.
- 2 days → 2,899 canceled, 5,573 not canceled.
- 4 days → 1,941 canceled, 3,952 not canceled.
- Short stays (1 day): 1,466 canceled, 5,138 not canceled → fewer cancellations relative to total bookings.
- Long stays (≥5 days): Cancellation rates vary but absolute numbers are lower due to fewer bookings.

Observation:

Medium-length stays (2–4 days) account for most cancellations, reflecting high booking volume in this segment.

Very long stays (10+ days) are rare; cancellation impact is minimal in these segments.

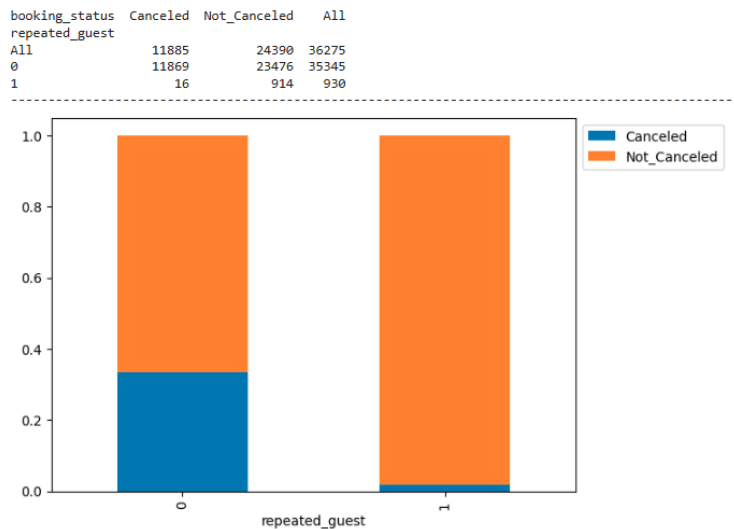


Figure 33: Repeated Guests vs Booking Status

Cancellations vs. Repeated Guests

- Overall: 11,885 bookings canceled, 24,390 not canceled.
- First-time guests (0 repeated): 11,869 canceled, 23,476 not canceled → Vast majority of cancellations occur among first-time guests.
- Repeated guests (1 or more bookings): 16 canceled, 914 not canceled → loyal guests rarely cancel.

Observation:

- Guest loyalty strongly reduces cancellation risk.
- Encouraging repeat bookings through loyalty programs or personalized incentives could help lower overall cancellations.

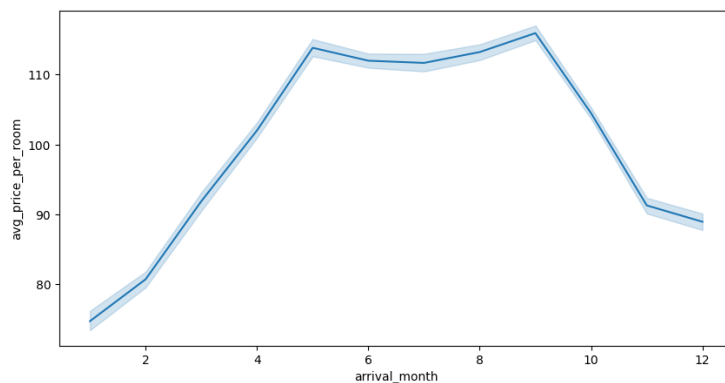


Figure 34: Arrival Month vs Avg. Price per Room

The booking trend indicates that **guest numbers remain steady from April to July**, averaging between **3,000 and 3,500 bookings** per month.

- The **highest volume of bookings** occurs in **October**, exceeding **5,000 reservations**.
- The **lowest bookings** are observed in **January**, with approximately **1,000 reservations**.

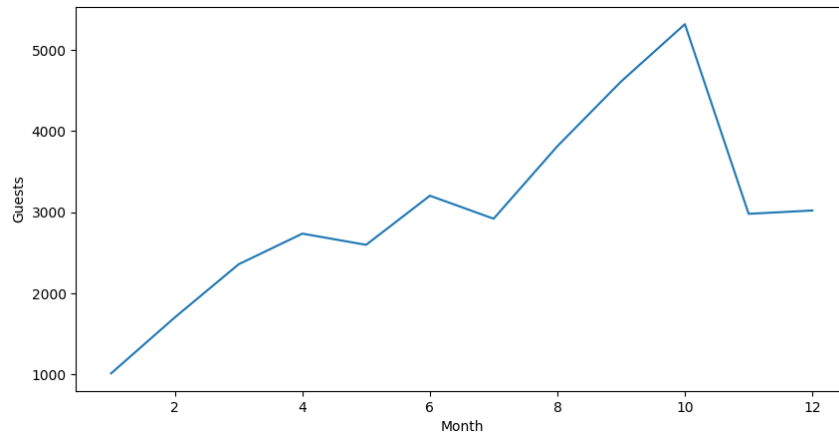


Figure 35: Month vs Guests

The price of rooms is highest in May to September - around 115 euros per room.

booking_status	Canceled	Not_Canceled	All
arrival_month			
All	11885	24390	36275
10	1880	3437	5317
9	1538	3073	4611
8	1488	2325	3813
7	1314	1606	2920
6	1291	1912	3203
4	995	1741	2736
5	948	1650	2598
11	875	2105	2980
3	700	1658	2358
2	430	1274	1704
12	402	2619	3021
1	24	990	1014

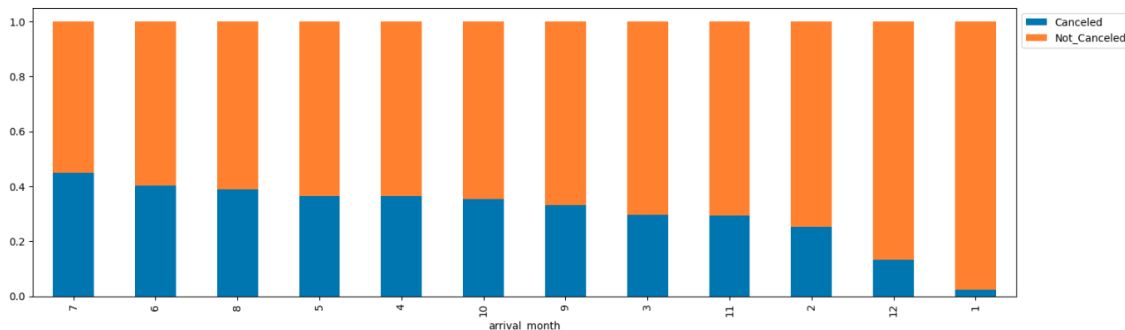


Figure 36: Arrival Month vs Booking Status

Cancellations vs. Arrival Month

- Overall: 11,885 bookings canceled, 24,390 not canceled.
- Peak cancellations:
 - October: 1,880 canceled, 3,437 not canceled.
 - September: 1,538 canceled, 3,073 not canceled.
 - August: 1,488 canceled, 2,325 not canceled.
- Lowest cancellations:
 - January: 24 canceled, 990 not canceled.
 - February: 430 canceled, 1,274 not canceled.

Observation: Cancellations are higher during peak season months (Aug–Oct), reflecting higher booking volumes. Winter months (Jan–Feb) have low cancellation numbers, consistent with lower occupancy.

EDA Questions

Q1: What are the busiest months in the hotel?

The busiest months are **October, September, and August.**

October accounts for the highest share with 14.7% of total bookings.

September follows with 12.7%, and August with 10.5%.

These months experience peak demand due to factors such as holidays, seasonal events, or favorable weather. The hotel should ensure adequate resources and staffing during these periods to maintain service quality.

Q2: Which market segment do most of the guests come from?

The majority of guests belong to the online market segment, which represents 64% of total bookings. This highlights the dominance of online channels such as hotel websites, OTAs, and direct online reservations.

The offline segment contributes about 29% of bookings, which may come from walk-ins, phone reservations, or traditional travel agencies. This insight emphasizes the need for focused marketing efforts on online platforms while maintaining offline relationships.

Q3: Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?

Room prices differ across market segments. The online segment records the highest average prices, likely due to the convenience of online booking platforms.

In contrast, the offline segment shows comparatively lower rates, possibly because of direct negotiations or discounts offered through traditional travel agents.

These differences may also be influenced by customer demographics, booking behaviors, and seasonal demand. The hotel should continuously adjust its pricing strategy to maximize revenue across all segments.

Q4: What percentage of bookings are canceled?

Approximately 32.8% of all bookings are canceled.

Q5: Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?

Around 2.6% of total bookings (930 reservations) are from repeating guests. Out of these, only 16 were canceled, resulting in a cancellation rate of just 1.7%.

This indicates that repeating guests demonstrate high commitment and are far less likely to cancel compared to first-time guests.

Q6: Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?

Yes, special requests significantly affect cancellations. As the number of special requests increases, cancellations decrease:

No special requests → 43.22% cancellations

1 request → 23.79% cancellations

2 requests → 14.61% cancellations

3–5 requests → 0% cancellations

This trend suggests that guests with specific requirements are more engaged and less likely to cancel, making them more reliable customers.

Data Preprocessing

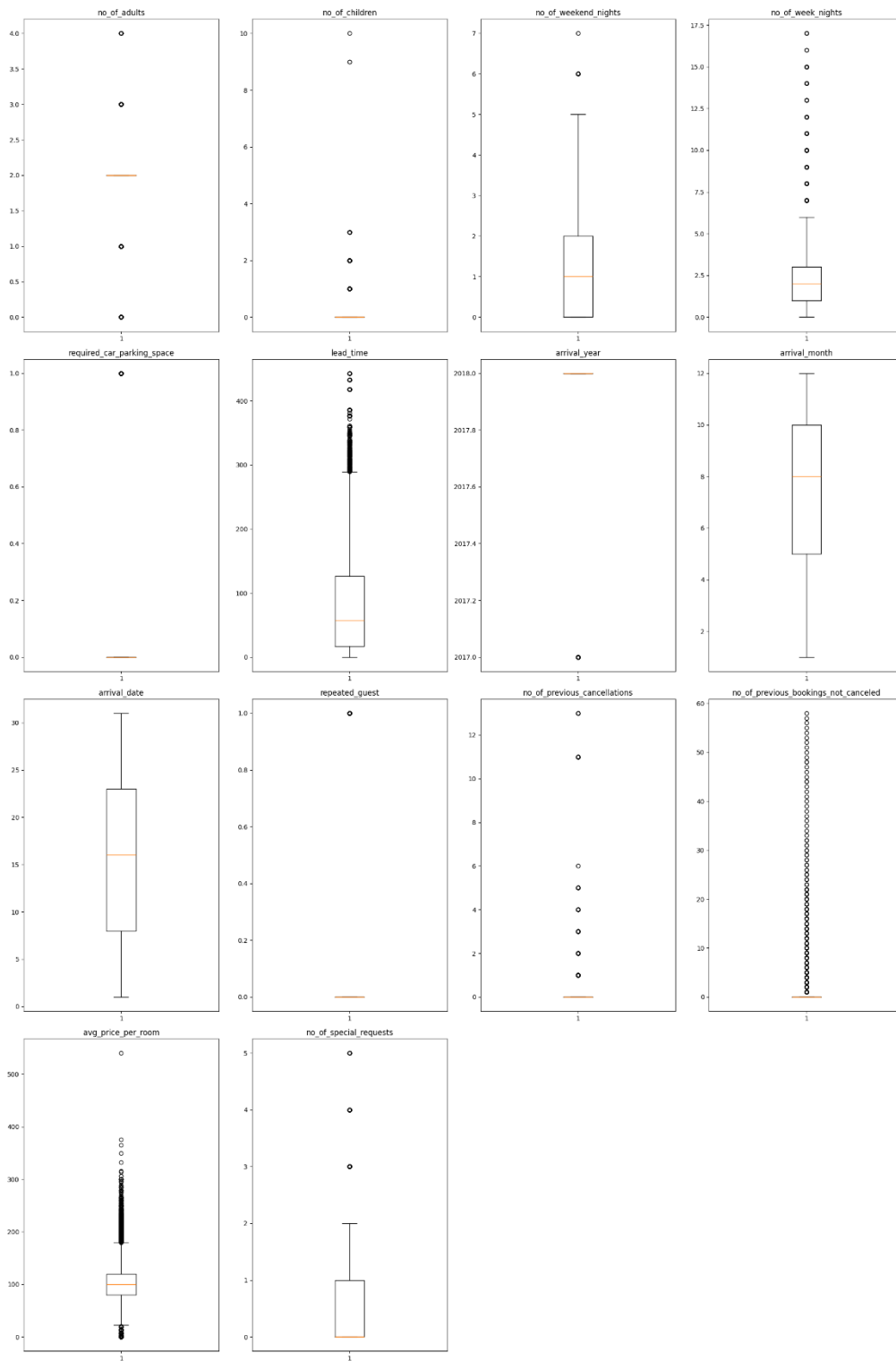


Figure 37: Outlier detection and Treatment

Treatment of Outliers

Decision: Outliers are retained in the analysis.

Rationale: They contain valuable information, offering insights into extreme booking behaviors, high-value transactions, or unique guest preferences.

Observation: Including outliers ensures a more comprehensive understanding of the data, supporting better-informed business decisions.

Model Building

Data Preparation for Logistic Regression

```
Number of rows in train data = 25392
Number of rows in test data = 10883
Percentage of classes in training set:
booking_status
0    0.670644
1    0.329356
Name: proportion, dtype: float64
Percentage of classes in test set:
booking_status
0    0.676376
1    0.323624
Name: proportion, dtype: float64
```

Figure 38: Data Preparation for Logistic Regression

Building Logistic Regression Model

```
Warning: Maximum number of iterations has been exceeded.
Current function value: 0.424975
Iterations: 35

=====
Logit Regression Results
=====
Dep. Variable:    booking_status    No. Observations:    25392
Model:            Logit            DF Residuals:        25364
Method:           MLE              DF Model:            27
Date:            Sun, 31 Aug 2025    Pseudo R-squ.:      0.3294
Time:            05:59:46           Log-Likelihood:     -10791.
converged:        False            LL-Null:            -16091.
Covariance Type:  nonrobust         LLR p-value:        0.000

=====
              coef    std err          z      P>|z|    [0.025    0.975]
-----
const                -923.3027    120.812    -7.642    0.000   -1160.090   -686.516
no_of_adults           0.1143     0.038     3.038    0.002     0.041     0.188
no_of_children         0.1562     0.057     2.729    0.006     0.044     0.268
no_of_weekend_nights   0.1075     0.020     5.435    0.000     0.069     0.146
no_of_week_nights      0.0404     0.012     3.285    0.001     0.016     0.064
required_car_parking_space -1.5938     0.138   -11.561    0.000     -1.864    -1.324
lead_time              0.0157     0.000    58.872    0.000     0.015     0.016
arrival_year           0.4563     0.060     7.622    0.000     0.339     0.574
arrival_month          -0.0416     0.006    -6.424    0.000     -0.054    -0.029
arrival_date           0.0005     0.002     0.253    0.800     -0.003     0.004
repeated_guest         -2.3478     0.617    -3.807    0.000     -3.557    -1.139
no_of_previous_cancellations 0.2665     0.086     3.108    0.002     0.098     0.435
no_of_previous_bookings_not_canceled -0.1726     0.152    -1.132    0.258     -0.471     0.126
avg_price_per_room      0.0188     0.001    25.462    0.000     0.017     0.020
no_of_special_requests -1.4689     0.030   -48.788    0.000     -1.528    -1.410
type_of_meal_plan_Meal Plan 2 0.1760     0.067     2.641    0.008     0.045     0.307
type_of_meal_plan_Meal Plan 3 17.8312    5046.854     0.004    0.997   -9873.820   9909.482
type_of_meal_plan_Not Selected 0.2769     0.053     5.223    0.000     0.173     0.381
room_type_reserved_Room_Type 2 -0.3587     0.131    -2.743    0.006     -0.615    -0.102
room_type_reserved_Room_Type 3 7.312e-05    1.310    5.58e-05    1.000     -2.567     2.567
room_type_reserved_Room_Type 4 -0.2822     0.053    -5.308    0.000     -0.386    -0.178
room_type_reserved_Room_Type 5 -0.7162     0.209    -3.425    0.001     -1.126    -0.306
room_type_reserved_Room_Type 6 -0.9457     0.147    -6.436    0.000     -1.234    -0.658
room_type_reserved_Room_Type 7 -1.3961     0.293    -4.767    0.000     -1.970    -0.822
market_segment_type_Complementary -41.8633    8.38e+05   -5e-05    1.000   -1.64e+06   1.64e+06
market_segment_type_Corporate -1.1922     0.266    -4.481    0.000     -1.714    -0.671
market_segment_type_Offline -2.1940     0.255    -8.617    0.000     -2.693    -1.695
market_segment_type_Online -0.3929     0.251    -1.564    0.118     -0.885     0.100
=====
```

Figure 39: Summary 1

Key Drivers of Booking Cancellations

Positive impact on cancellations (increase likelihood):

- Number of adults & children: Slightly higher likelihood of cancellation as group size increases.
- Weekend and week nights: Longer stays marginally increase cancellation probability.
- Lead time: Bookings made further in advance are more likely to be canceled.
- Previous cancellations: Guests with prior cancellations are more likely to cancel again.
- Meal Plan 2 & Not Selected: Slightly higher cancellation likelihood compared to Meal Plan 1.
- Average price per room: Higher-priced bookings are somewhat more likely to be canceled.

Negative impact on cancellations (reduce likelihood):

- Required car parking space: Guests requiring parking are less likely to cancel.
- Repeated guests: Loyal guests are significantly less likely to cancel.
- Special requests: More special requests strongly reduce cancellation probability.
- Room types 2–7: Booking premium or specific room types lowers cancellation risk.
- Corporate and Offline market segments: These bookings are less likely to cancel compared to the baseline.
- Insignificant or unreliable variables:
Certain categories (e.g., Meal Plan 3, Room Type 3, Complementary market segment) have extremely high standard errors or p-values ≈ 1 , suggesting insufficient data or unstable estimates.

Observation:

- Key predictors: Lead time, repeated guests, special requests, and market segment clearly influence cancellations.
- Management takeaway: Strategies such as targeting first-time guests, long lead bookings, and online bookings for cancellation mitigation are supported by this model.

Model Performance & Reliability

Training performance Logistic Regression Model:				
	Accuracy	Recall	Precision	F1
0	0.805923	0.634103	0.739506	0.68276

Figure 40: Training Performance LG Model 1

- F1 Score: $\sim 0.68 \rightarrow$ The model shows moderate predictive performance for booking cancellations. Efforts will be made to improve accuracy and reliability.
- Multicollinearity: Some variables may be highly correlated, which can distort p-values and coefficients in the logistic regression.
- Next Steps: Removing or addressing multicollinearity will allow the model to produce more reliable estimates and better inform management decisions.

Observation:

While the model provides useful insights, further refinement is needed to ensure actionable recommendations for cancellation mitigation and revenue management.

Testing the assumptions of linear regression model

Multicollinearity occurs when predictor variables in a regression model are correlated. This correlation is a problem because predictor variables should be independent. If the correlation between variables is high, it can cause problems when we fit the model and interpret the results. When we have multicollinearity in the linear model, the coefficients that the model suggests are unreliable.

There are different ways of detecting (or testing) multicollinearity. One such way is by using the Variance Inflation Factor, or VIF.

Variance Inflation Factor (VIF): Variance inflation factors measure the inflation in the variances of the regression parameter estimates due to collinearities that exist among the predictors. It is a measure of how much the variance of the estimated regression coefficient β_k is "inflated" by the existence of correlation among the predictor variables in the model.

If VIF is 1, then there is no correlation among the k th predictor and the remaining predictor variables, and hence, the variance of β_k is not inflated at all.

General Rule of thumb:

If VIF is between 1 and 5, then there is low multicollinearity.

If VIF is between 5 and 10, we say there is moderate multicollinearity.

If VIF is exceeding 10, it shows signs of high multicollinearity.

Checking Multicollinearity

VIF before removing variables:

	Features	VIF
0	const	39468156.70600
1	no_of_adults	1.34815
2	no_of_children	1.97823
3	no_of_weekend_nights	1.06948
4	no_of_week_nights	1.09567
5	required_car_parking_space	1.03993
6	lead_time	1.39491
7	arrival_year	1.43083
8	arrival_month	1.27567
9	arrival_date	1.00674
10	repeated_guest	1.78352
11	no_of_previous_cancellations	1.39569
12	no_of_previous_bookings_not_canceled	1.65199
13	avg_price_per_room	2.05042
14	no_of_special_requests	1.24728
15	type_of_meal_plan_Meal Plan 2	1.27185
16	type_of_meal_plan_Meal Plan 3	1.02522
17	type_of_meal_plan_Not Selected	1.27218
18	room_type_reserved_Room_Type 2	1.10144
19	room_type_reserved_Room_Type 3	1.00330
20	room_type_reserved_Room_Type 4	1.36152
21	room_type_reserved_Room_Type 5	1.02781
22	room_type_reserved_Room_Type 6	1.97307
23	room_type_reserved_Room_Type 7	1.11512
24	market_segment_type_Complementary	4.50011
25	market_segment_type_Corporate	16.92844
26	market_segment_type_Offline	64.11392
27	market_segment_type_Online	71.17643

Figure 41: VIF Before Removing Variables

	Features	VIF
0	const	39468156.70600
25	market_segment_type_Corporate	16.92844
26	market_segment_type_Offline	64.11392
27	market_segment_type_Online	71.17643

Figure 42: Multicollinearity Analysis

Multicollinearity Analysis

High VIF values:

- const → 39,468,156 (expected for intercept)
- market_segment_type_Corporate → 16.93
- market_segment_type_Offline → 64.11
- market_segment_type_Online → 71.18

Observation:

- The market segment variables exhibit strong multicollinearity, which can distort regression coefficients and p-values.
- Next Steps: Consider removing or combining highly correlated variables to improve the reliability of the model and make coefficients more interpretable.

VIF after dropping market_segment_type_Online

	Features	VIF
0	const	39391371.31459
1	no_of_adults	1.33178
2	no_of_children	1.97735
3	no_of_weekend_nights	1.06904
4	no_of_week_nights	1.09512
5	required_car_parking_space	1.03979
6	lead_time	1.39064
7	arrival_year	1.42838
8	arrival_month	1.27463
9	arrival_date	1.00672
10	repeated_guest	1.78019
11	no_of_previous_cancellations	1.39545
12	no_of_previous_bookings_not_canceled	1.65175
13	avg_price_per_room	2.04959
14	no_of_special_requests	1.24242
15	type_of_meal_plan_Meal Plan 2	1.27150
16	type_of_meal_plan_Meal Plan 3	1.02522
17	type_of_meal_plan_Not Selected	1.27039
18	room_type_reserved_Room_Type 2	1.10127
19	room_type_reserved_Room_Type 3	1.00330
20	room_type_reserved_Room_Type 4	1.35600
21	room_type_reserved_Room_Type 5	1.02781
22	room_type_reserved_Room_Type 6	1.97273
23	room_type_reserved_Room_Type 7	1.11500
24	market_segment_type_Complementary	1.33825
25	market_segment_type_Corporate	1.52777
26	market_segment_type_Offline	1.59742

Figure 43: Revised VIF values 1

Dealing with high p-value variables

- Some of the dummy variables in the data have p-value > 0.05. So, they are not significant and we'll drop them
- But sometimes p-values change after dropping a variable. So, we'll not drop all variables at once
- Instead:
 - Build a model, check the p-values of the variables, and drop the column with the highest p-value
 - Create a new model without the dropped feature, check the p-values of the variables, and drop the column with the highest p-value
 - Repeat the above two steps till there are no columns with p-value > 0.05

Note: The above process can also be done manually by picking one variable at a time that has a high p-value, dropping it, and building a model again. But that might be a little tedious and using a loop will be more efficient.

Optimization terminated successfully.

Current function value: 0.425677

Iterations 11

Logit Regression Results

Dep. Variable:	booking_status	No. Observations:	25392
Model:	Logit	Df Residuals:	25370
Method:	MLE	Df Model:	21
Date:	Sun, 31 Aug 2025	Pseudo R-squ.:	0.3283
Time:	06:00:51	Log-Likelihood:	-10809.
converged:	True	LL-Null:	-16091.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P> z	[0.025	0.975]
const	-917.2860	120.456	-7.615	0.000	-1153.376	-681.196
no_of_adults	0.1086	0.037	2.914	0.004	0.036	0.182
no_of_children	0.1522	0.057	2.660	0.008	0.040	0.264
no_of_weekend_nights	0.1086	0.020	5.501	0.000	0.070	0.147
no_of_week_nights	0.0418	0.012	3.403	0.001	0.018	0.066
required_car_parking_space	-1.5943	0.138	-11.561	0.000	-1.865	-1.324
lead_time	0.0157	0.000	59.218	0.000	0.015	0.016
arrival_year	0.4531	0.060	7.591	0.000	0.336	0.570
arrival_month	-0.0424	0.006	-6.568	0.000	-0.055	-0.030
repeated_guest	-2.7365	0.557	-4.915	0.000	-3.828	-1.645
no_of_previous_cancellations	0.2289	0.077	2.983	0.003	0.078	0.379
avg_price_per_room	0.0192	0.001	26.343	0.000	0.018	0.021
no_of_special_requests	-1.4699	0.030	-48.892	0.000	-1.529	-1.411
type_of_meal_plan_Meal Plan 2	0.1654	0.067	2.487	0.013	0.035	0.296
type_of_meal_plan_Not Selected	0.2858	0.053	5.405	0.000	0.182	0.389
room_type_reserved_Room_Type 2	-0.3560	0.131	-2.725	0.006	-0.612	-0.100
room_type_reserved_Room_Type 4	-0.2826	0.053	-5.330	0.000	-0.387	-0.179
room_type_reserved_Room_Type 5	-0.7352	0.208	-3.529	0.000	-1.143	-0.327
room_type_reserved_Room_Type 6	-0.9650	0.147	-6.572	0.000	-1.253	-0.677
room_type_reserved_Room_Type 7	-1.4312	0.293	-4.892	0.000	-2.005	-0.858
market_segment_type_Corporate	-0.7928	0.103	-7.711	0.000	-0.994	-0.591
market_segment_type_Offline	-1.7867	0.052	-34.391	0.000	-1.889	-1.685

Figure 44: Summary 2

VIF values:

```
const                39193128.05210
no_of_adults         1.33089
no_of_children       1.97571
no_of_weekend_nights 1.06048
no_of_week_nights    1.08993
required_car_parking_space 1.03921
lead_time            1.37597
arrival_year         1.42115
arrival_month        1.26698
repeated_guest       1.53128
no_of_previous_cancellations 1.20390
avg_price_per_room   1.72339
no_of_special_requests 1.23777
type_of_meal_plan_Meal Plan 2 1.25776
type_of_meal_plan_Not Selected 1.25153
room_type_reserved_Room_Type 2 1.10085
room_type_reserved_Room_Type 4 1.34479
room_type_reserved_Room_Type 5 1.01759
room_type_reserved_Room_Type 6 1.94361
room_type_reserved_Room_Type 7 1.05823
market_segment_type_Corporate 1.45290
market_segment_type_Offline 1.55675
dtype: float64
```

Figure 45: VIF_Values 2

Model performance evaluation

Training performance model 1:

	Accuracy	Recall	Precision	F1
0	0.80592	0.63410	0.73951	0.68276

Figure 46: Training Performance Model 1

Training performance model 2:

	Accuracy	Recall	Precision	F1
0	0.80541	0.63255	0.73903	0.68166

Figure 47: Training Performance Model 2

Both models perform similarly, achieving moderate predictive performance.

F1 score (~0.68) indicates a balanced trade-off between recall and precision, suitable for identifying cancellations.

Further improvement can be achieved by refining features, addressing multicollinearity, and optimizing the model.

Interpreting Coefficients as Odds

Logistic regression coefficients can be transformed into odds ratios using the exponential function:

Odds Ratio = $e^{\text{Coefficient}}$ Odds Ratio = $e^{\text{Coefficient}}$

Interpretation:

Odds Ratio > 1: The variable increases the likelihood of cancellation.

Odds Ratio < 1: The variable decreases the likelihood of cancellation.

Odds Ratio = 1: The variable has no effect on cancellation probability.

Example:

A coefficient of 0.114 for no_of_adults → Odds ratio $e^{0.114} \approx 1.12$ $e^{0.114} \approx 1.12$

Interpretation: Each additional adult increases the odds of cancellation by ~12%.

A coefficient of -1.4689 for no_of_special_requests → Odds ratio $e^{-1.4689} \approx 0.23$ $e^{-1.4689} \approx 0.23$

Interpretation: Each additional special request reduces the odds of cancellation by ~77%.

Business Insight:

Converting coefficients to odds makes it easier for management to understand the impact of each factor on cancellations and prioritize mitigation strategies.

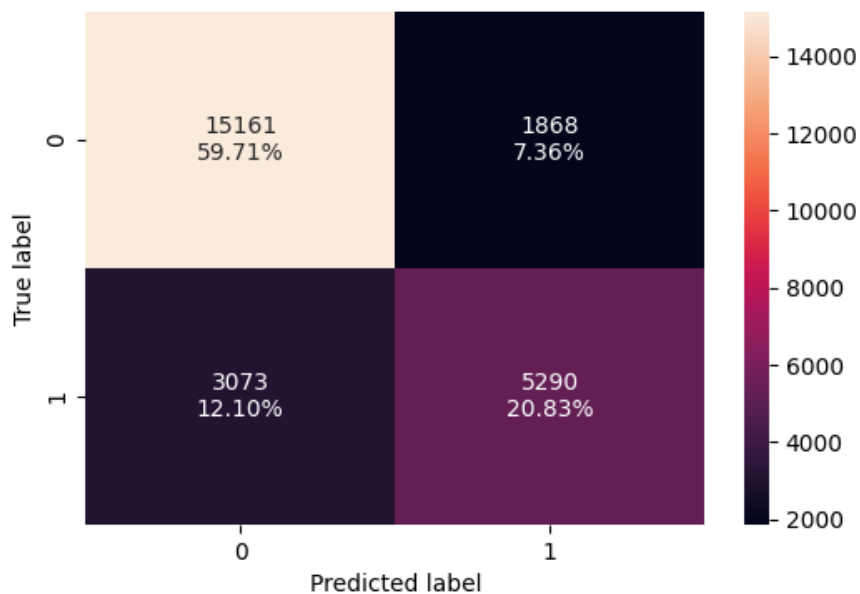


Figure 48: Confusion Matrix 1

Training performance:

	Accuracy	Recall	Precision	F1
0	0.80541	0.63255	0.73903	0.68166

Figure 49: Training Performance 1

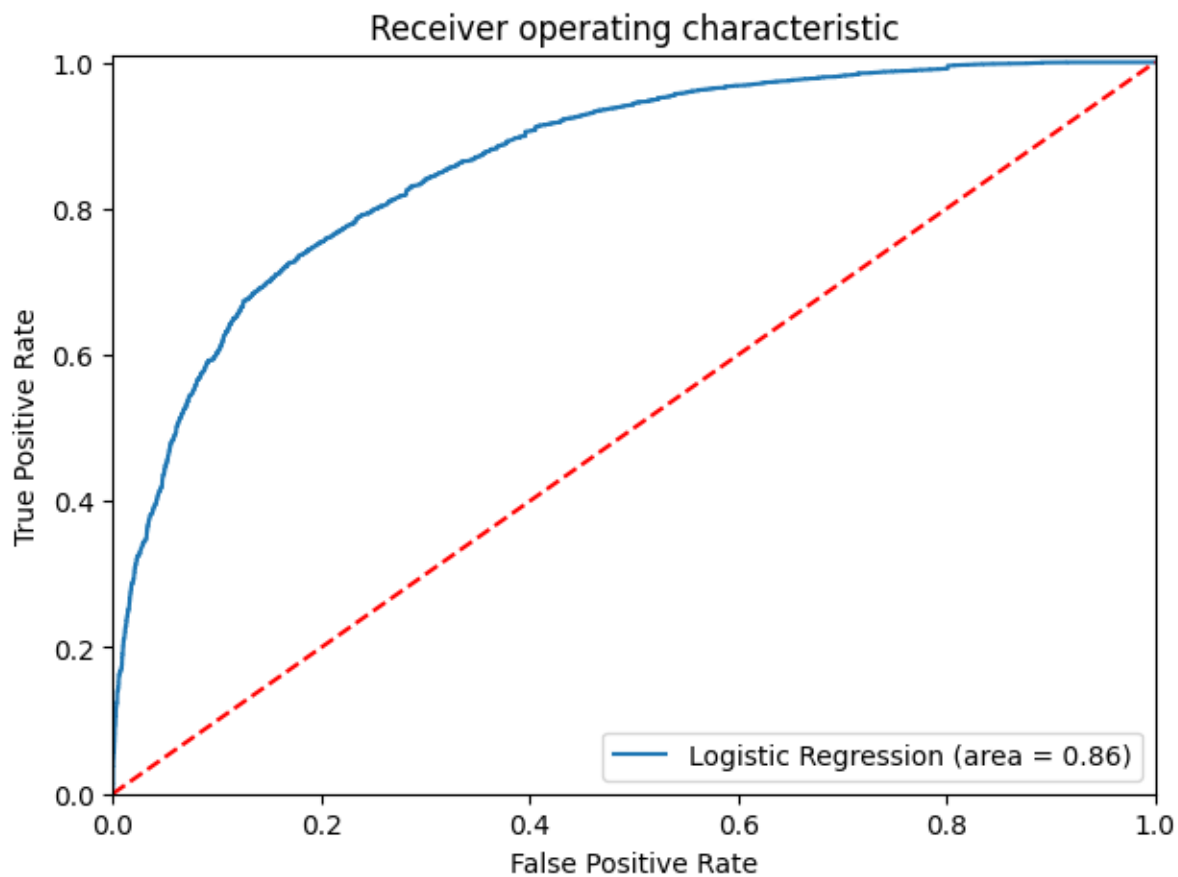


Figure 50: ROC 1

The Logistic Regression model demonstrates a well-generalized performance across both the training and test sets.

The model achieves a ROC-AUC score of 0.86 on the training set, which indicates strong discriminatory power and reliable predictive capability.

Creating the confusion matrix for X_train1 and y_train with optimal_threshold_auc_roc as threshold

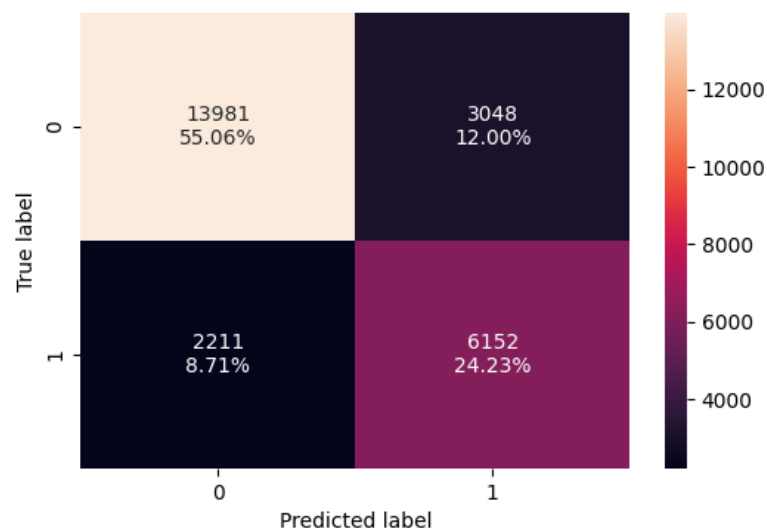


Figure 51: Confusion Matrix 2

Training performance:

	Accuracy	Recall	Precision	F1
0	0.79289	0.73562	0.66870	0.70056

Figure 52: Training Performance 2

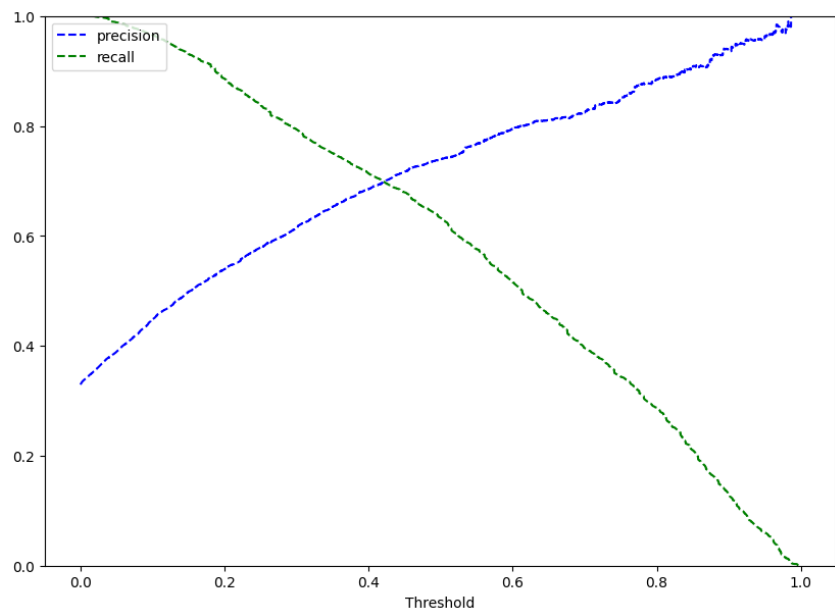


Figure 53: Precisions, Recalls, Thresholds

Setting The Threshold: 0.42

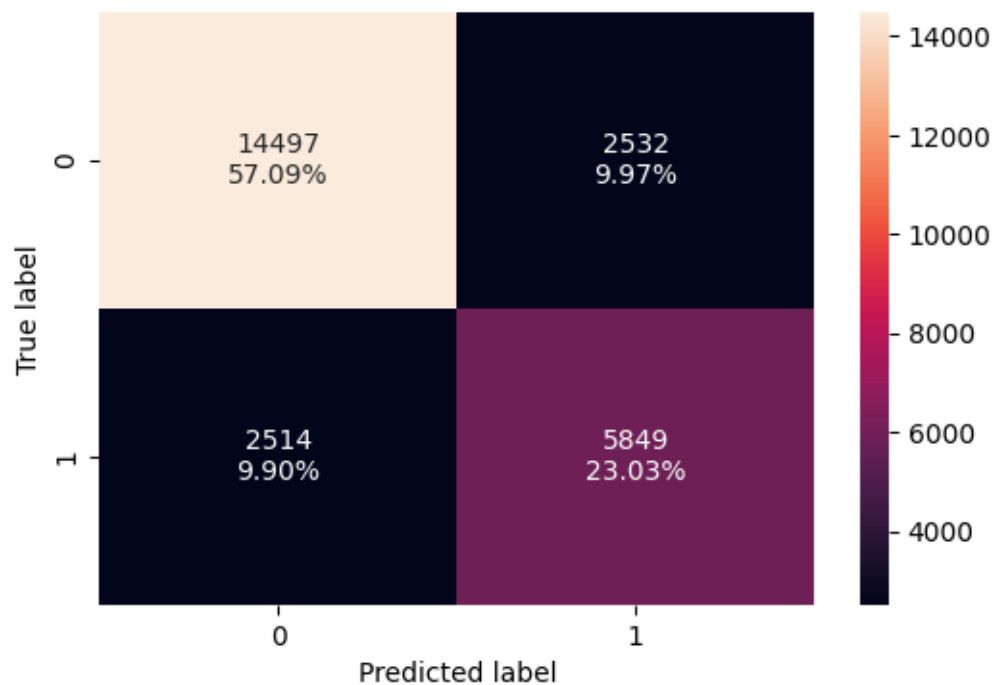


Figure 54: Confusion Matrix 3

Training performance:

	Accuracy	Recall	Precision	F1
0	0.80128	0.69939	0.69789	0.69864

Figure 55: Training Performance 3

- The updated model shows slightly improved recall and F1 score compared to previous versions.
- Indicates better ability to identify cancellations while maintaining balanced precision.
- Performance is moderate, with room for improvement through feature refinement and handling multicollinearity.

Let's check the performance on the test set

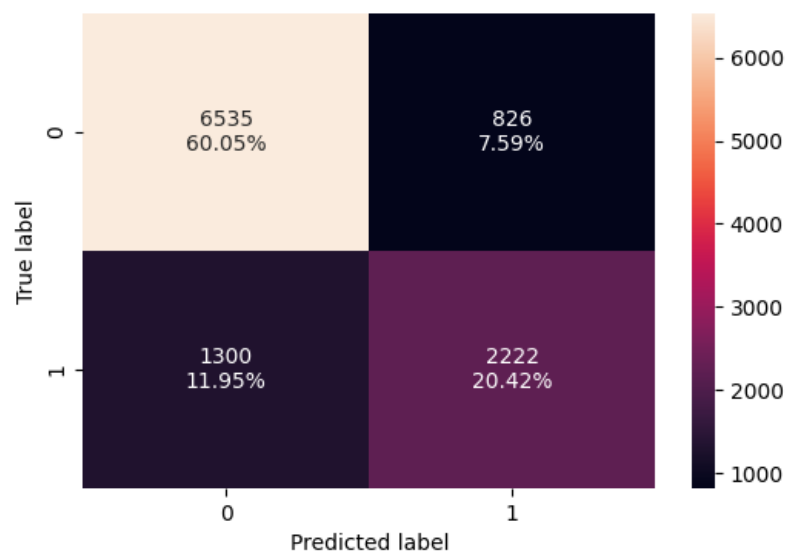


Figure 56: Confusion Matrix 4

Test performance:

	Accuracy	Recall	Precision	F1
0	0.80465	0.63089	0.72900	0.67641

Figure 57: Test Performance 1

- The model performs similarly on test data as on training data, indicating good generalization.
- F1 score (~0.68) suggests a balanced performance in identifying cancellations while controlling false positives.
- Slightly lower recall indicates some missed cancellations, which could be improved with feature optimization or handling multicollinearity.

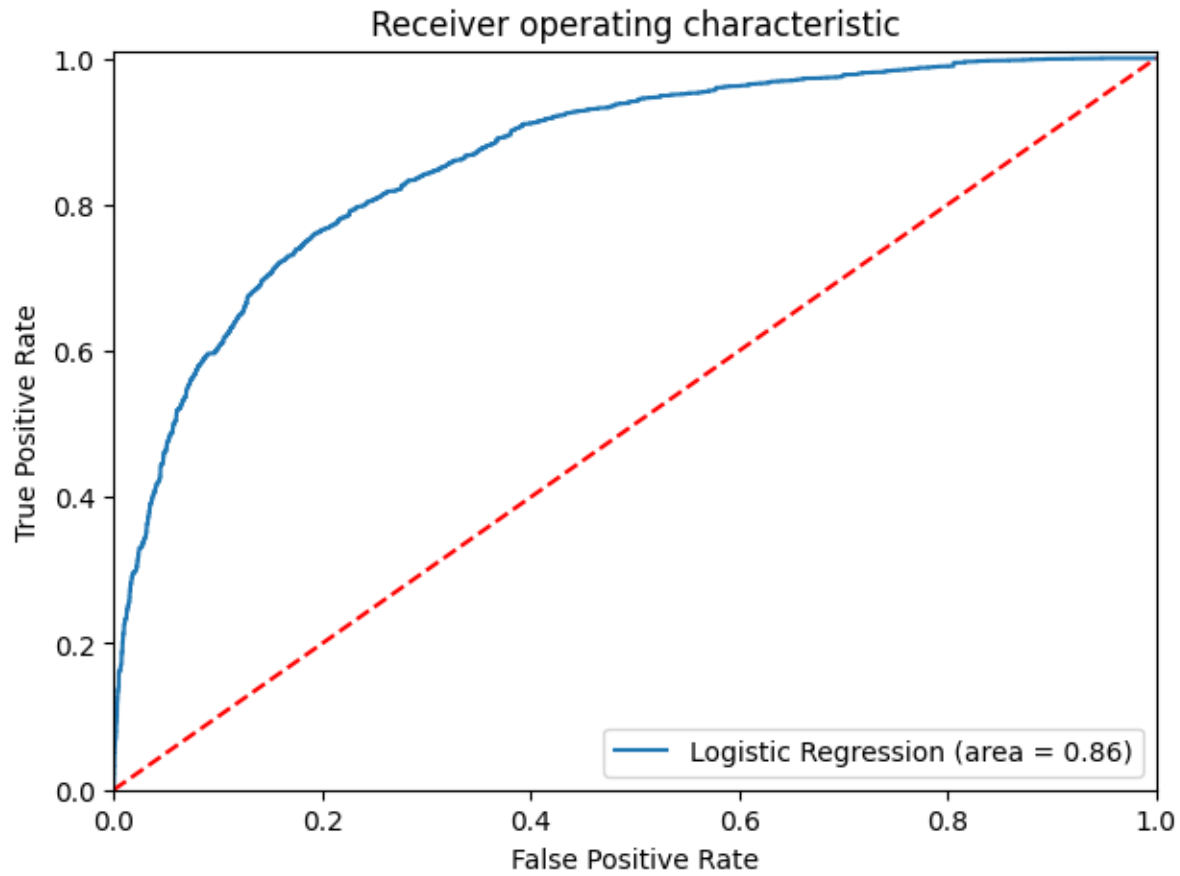


Figure 58: ROC 2

- AUC close to 1: Excellent discrimination.
- AUC ≈ 0.5 : No better than random guessing.
- The plotted curve shows the model performance visually, with the diagonal red line representing random chance.
- Observation: The logistic regression model shows good discriminatory power, confirming its utility for identifying likely cancellations and supporting targeted interventions.

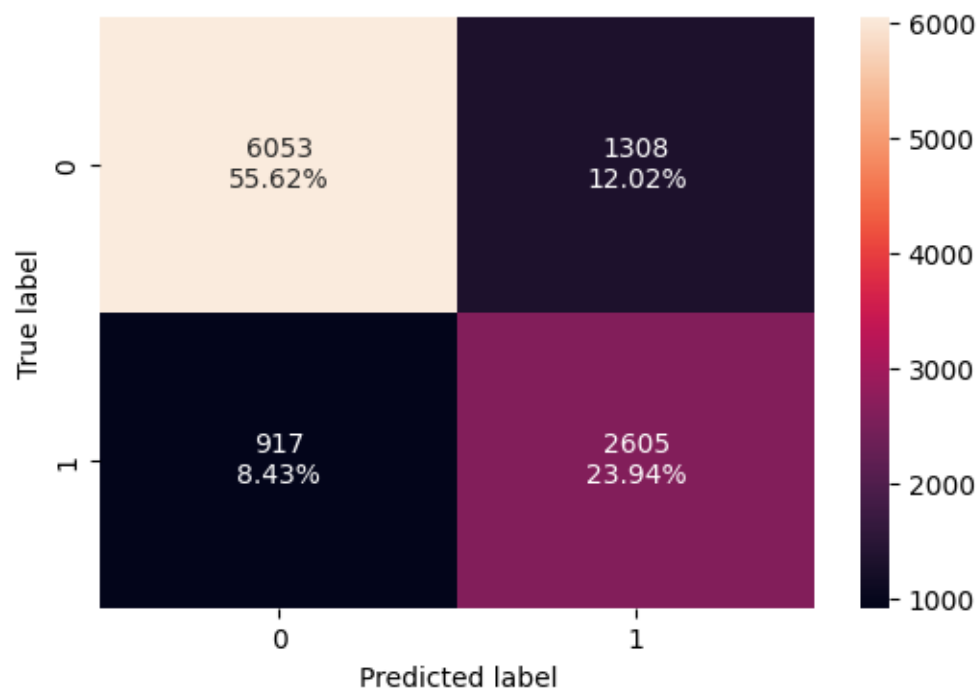


Figure 59: Confusion Matrix 5

Test performance:

	Accuracy	Recall	Precision	F1
0	0.79601	0.73935	0.66667	0.70113

Figure 60: Test Performance 2

- The model shows improved recall (~73.9%), meaning it can identify a larger portion of cancellations.
- Precision is slightly lower (~66.7%), indicating some false positives.
- F1 score (~70.1%) reflects a balanced trade-off between recall and precision.
- Overall, the model demonstrates moderate-to-good predictive performance on unseen test data.

Threshold: 0.42

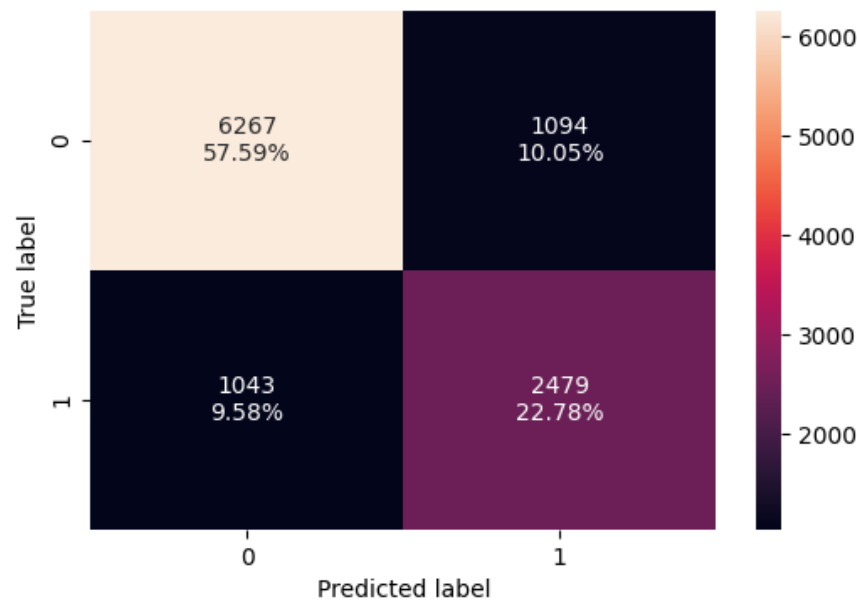


Figure 61: Confusion Matrix 6

Test performance:				
	Accuracy	Recall	Precision	F1
0	0.80364	0.70386	0.69381	0.69880

Figure 62: Test Performance 3

Training performance comparison

Training performance comparison:			
	Logistic Regression-default Threshold	Logistic Regression-0.37 Threshold	Logistic Regression-0.42 Threshold
Accuracy	0.80541	0.79289	0.80128
Recall	0.63255	0.73562	0.69939
Precision	0.73903	0.66870	0.69789
F1	0.68166	0.70056	0.69864

Figure 63: Training performance comparison

- Lowering the threshold (0.37, 0.42) increases recall, improving the model's ability to detect cancellations.
- Precision decreases slightly, showing more false positives.
- F1 score improves slightly with thresholds 0.37–0.42, indicating a better balance between recall and precision for cancellation prediction.
- Business Insight: Adjusting thresholds allows management to prioritize identifying likely cancellations while controlling false alarms.

Test Set performance comparison

Test set performance comparison:

	Logistic Regression-default Threshold (0.5)	Logistic Regression-0.37 Threshold	Logistic Regression-0.42 Threshold
Accuracy	0.80465	0.79601	0.80364
Recall	0.63089	0.73935	0.70386
Precision	0.72900	0.66667	0.69381
F1	0.67641	0.70113	0.69880

Figure 64: Test Set performance comparison

- Lowering the threshold (0.37 or 0.42) increases recall, enabling the model to detect more cancellations.
- Precision decreases slightly, reflecting more false positives.
- F1 score improves slightly with thresholds 0.37–0.42, offering a better balance between recall and precision.

Business Insight:

Adjusting thresholds can help the hotel proactively manage cancellations while balancing accuracy and alert volume.

Decision Tree model

Data Preparation for building Decision Tree

```
Shape of Training set : (25392, 27)
Shape of test set : (10883, 27)
Percentage of classes in training set:
booking_status
0    0.67064
1    0.32936
Name: proportion, dtype: float64
Percentage of classes in test set:
booking_status
0    0.67638
1    0.32362
Name: proportion, dtype: float64
```

Figure 65: Data Preparation for building Decision Tree

Checking model performance on training set

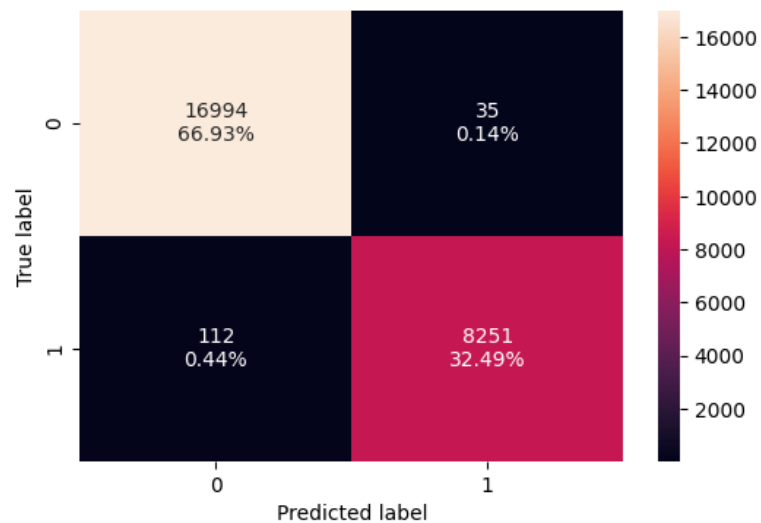


Figure 66: DT - CM 1

Training performance model0:

	Accuracy	Recall	Precision	F1
0	0.99421	0.98661	0.99578	0.99117

Figure 67: TPM 1

- The model shows exceptionally high performance on the training data.
- Both recall and precision are very high, indicating the model almost perfectly identifies cancellations and non-cancellations.
- Caution: Such high training metrics may indicate overfitting, so test set performance should be reviewed to ensure generalization to unseen data.

Checking model performance on test set

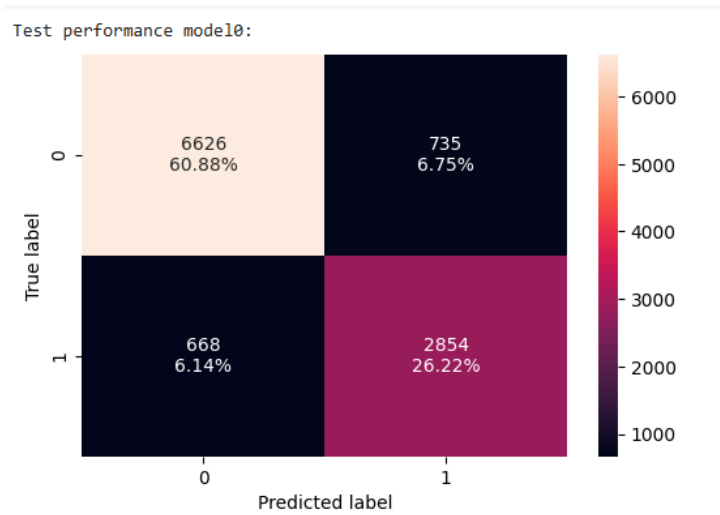


Figure 68: DT - CM 2

	Accuracy	Recall	Precision	F1
0	0.87108	0.81034	0.79521	0.80270

Figure 69: TPM 1 - Test

- The model shows strong performance, correctly predicting the majority of cancellations and non-cancellations.
- Recall (81%) indicates the model can detect most cancellations, while precision (79.5%) shows moderate false positives.
- F1 score (~80.3%) reflects a good balance between recall and precision, suitable for operational decision-making.

Before pruning the tree let's check the important features.

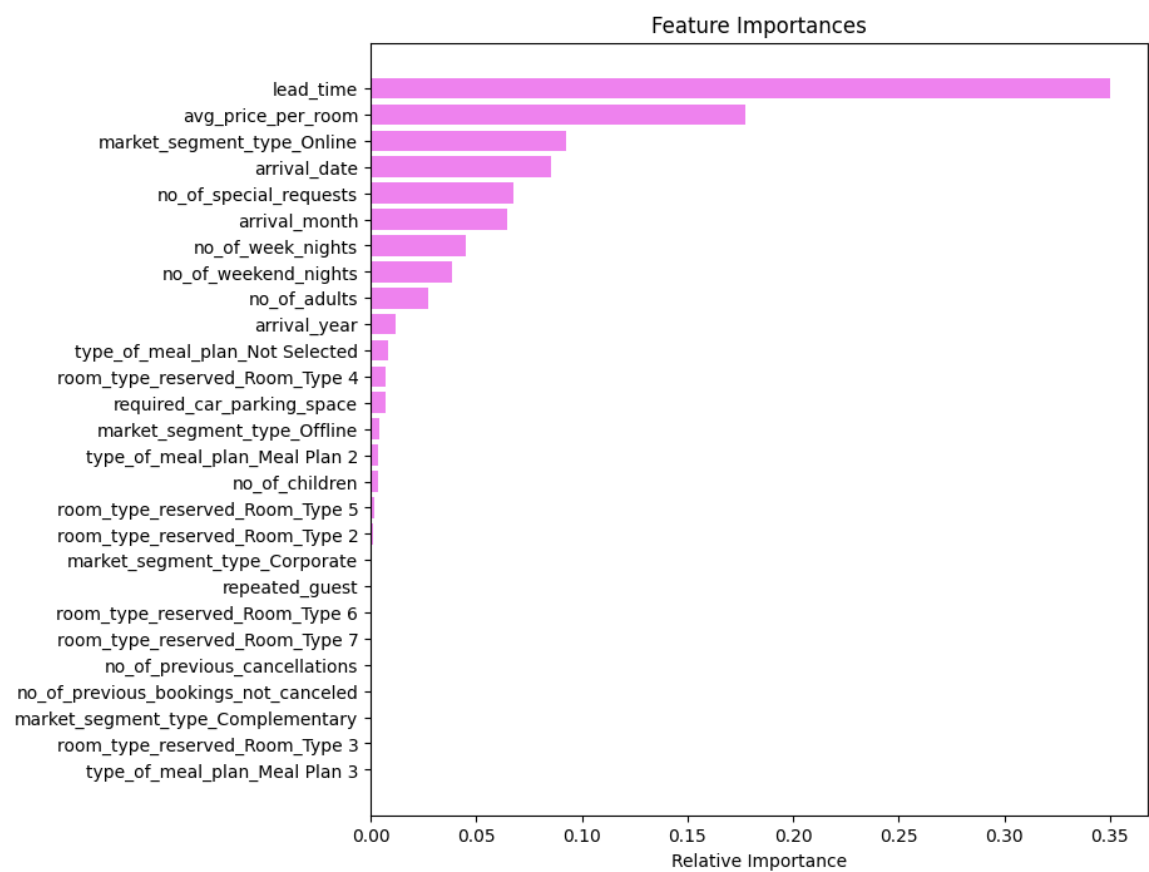


Figure 70: Feature Importance 1

Feature Importance Analysis

The bar chart shows the relative importance of each feature in predicting booking cancellations.

Key insights:

- Features with higher importance have the greatest influence on cancellation predictions.
- Features with lower importance contribute less and could potentially be simplified or removed.

Business Interpretation:

- Management can focus on high-impact factors (e.g., lead time, special requests, market segment) to develop targeted strategies for reducing cancellations.
- Insights from feature importance can guide resource allocation, promotions, and guest communication strategies.

Observation:

This analysis provides a clear visual hierarchy of predictive drivers, helping prioritize operational actions and interventions.

```
DecisionTreeClassifier
DecisionTreeClassifier(class_weight='balanced', max_depth=np.int64(6),
                      max_leaf_nodes=50, min_samples_split=10, random_state=1)
```

Figure 71: Decision Tree Classifier

Checking performance on training set

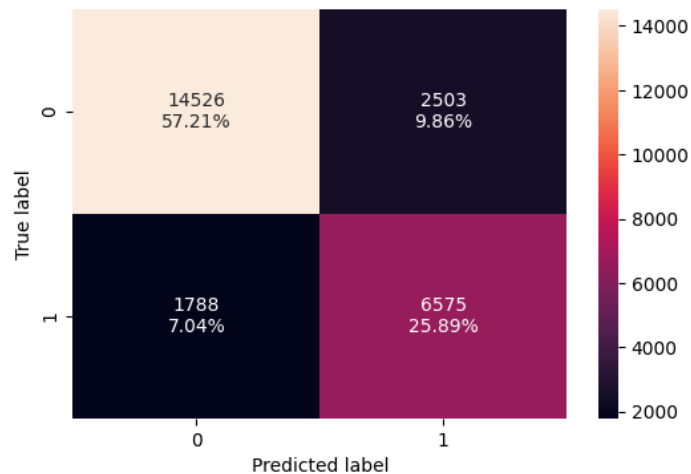


Figure 72: DT - CM 3

	Accuracy	Recall	Precision	F1
0	0.83101	0.78620	0.72428	0.75397

Figure 73: TPM 3 - Training

- The model shows good performance, capturing most cancellations while maintaining a reasonable level of false positives.
- Recall (~78.6%) indicates that a majority of cancellations are correctly identified.
- Precision (72.4%) indicates some false positives, but the F1 score (75.4%) suggests a balanced performance.
- This model can be used to proactively manage bookings and mitigate cancellation risk.

Checking performance on test set

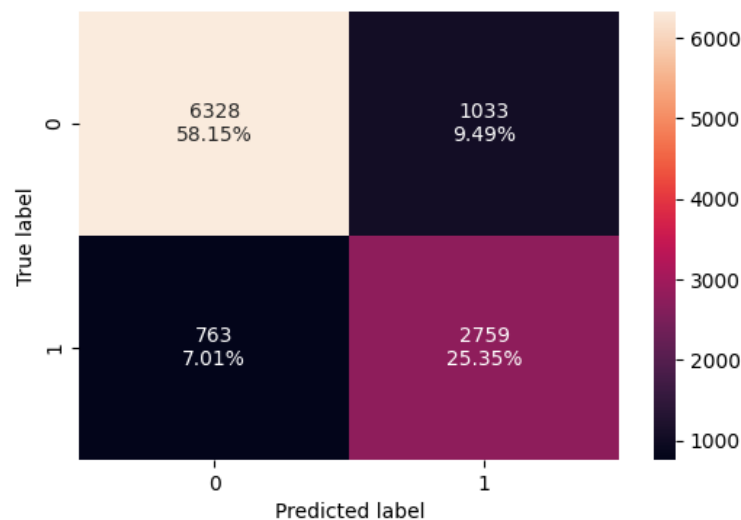


Figure 74: DT - CM 4

	Accuracy	Recall	Precision	F1
0	0.83497	0.78336	0.72758	0.75444

Figure 75: TPM 3 - Test

- The model demonstrates good predictive capability for identifying cancellations.
- Recall (~78.3%) shows the model captures a majority of cancellations.
- Precision (~72.8%) indicates some false positives, but overall F1 (~75.4%) reflects a balanced performance.
- Suitable for operational decision-making to mitigate booking cancellations effectively.

Visualizing the Decision Tree

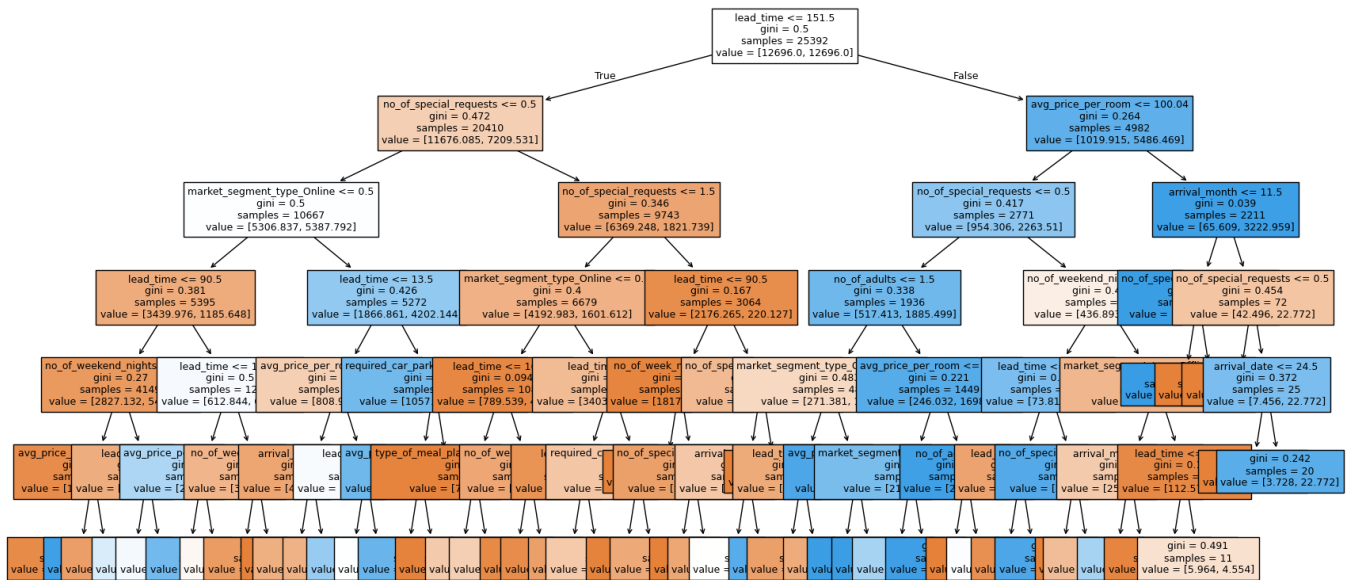


Figure 76: DT 1

- We can observe that the decision tree has become simpler, making the rules more interpretable and easier to understand.
- The model performance has been generalized after pruning.
- The most important features identified are:

Lead Time

- Market Segment – Online
- Number of Special Requests
- Average Price per Room

The rules derived from the decision tree can be interpreted as follows:

Lead time is the primary factor in determining whether a booking will be canceled, with 151 days chosen as the threshold for the first split.

For bookings made more than 151 days before the arrival date:

- If the average price per room is greater than 100 euros and the arrival month is December, the booking is less likely to be canceled.
- If the average price per room is less than or equal to 100 euros and the number of special requests is 0, the booking is more likely to be canceled.

For bookings made within 151 days of arrival:

- If a customer has at least 1 special request, the booking is less likely to be canceled.
- If the customer made no special requests:
- Bookings from the Online market segment are more likely to be canceled.
- Bookings from other market segments are less likely to be canceled.

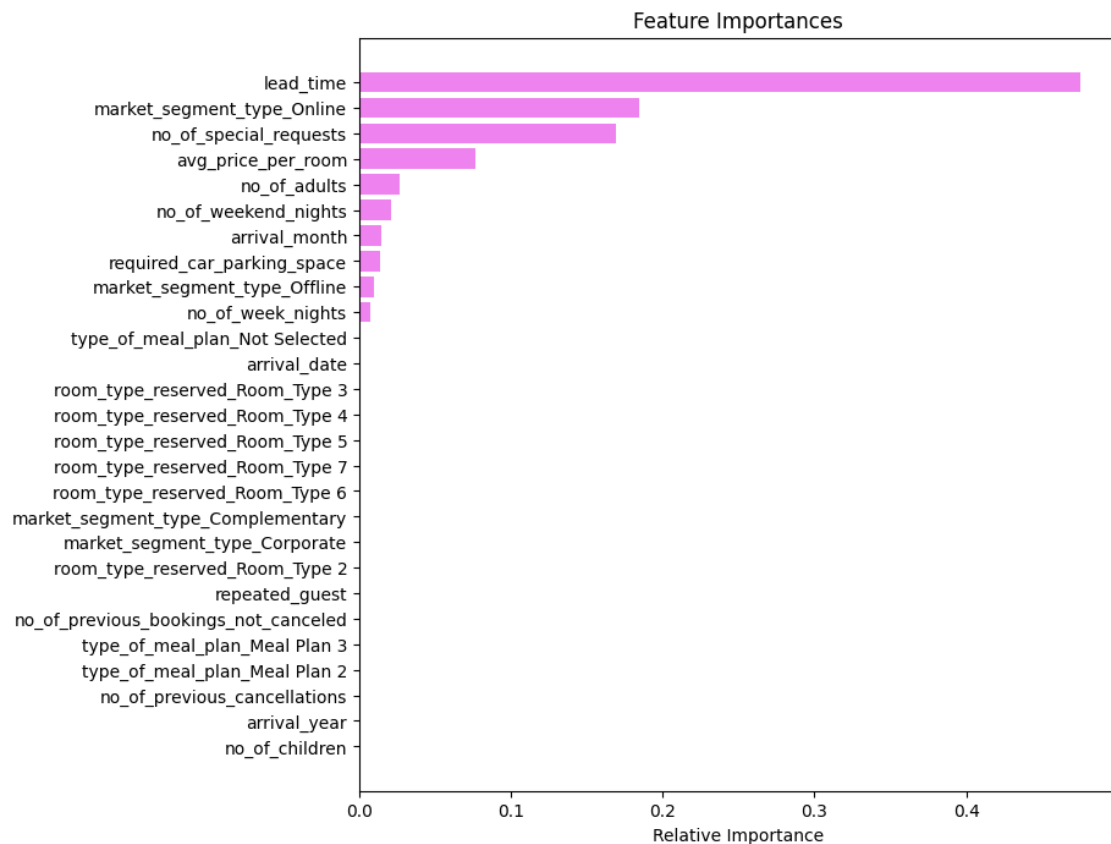


Figure 77: Feature Importance 2

Feature Importance in Tree-Based Model

The horizontal bar chart shows the relative importance of each feature in predicting booking cancellations.

Key insights:

Features with higher importance play a larger role in influencing cancellations.

Features with lower importance contribute less to the model's decisions.

Business Interpretation:

Focus on high-impact features such as lead time, special requests, number of adults/children, and market segment to develop strategies to reduce cancellations.

Helps prioritize operational interventions, promotional campaigns, or guest communications.

Observation:

This visualization provides a clear understanding of which factors drive cancellations, enabling management to make data-driven decisions to mitigate booking risks.

Cost Complexity Pruning

	ccp_alphas	impurities
0	0.00000	0.00838
1	-0.00000	0.00838
2	0.00000	0.00838
3	0.00000	0.00838
4	0.00000	0.00838
...	...	...
1837	0.00890	0.32806
1838	0.00980	0.33786
1839	0.01272	0.35058
1840	0.03412	0.41882
1841	0.08118	0.50000

1842 rows × 2 columns

Figure 78: Cost Complexity Pruning

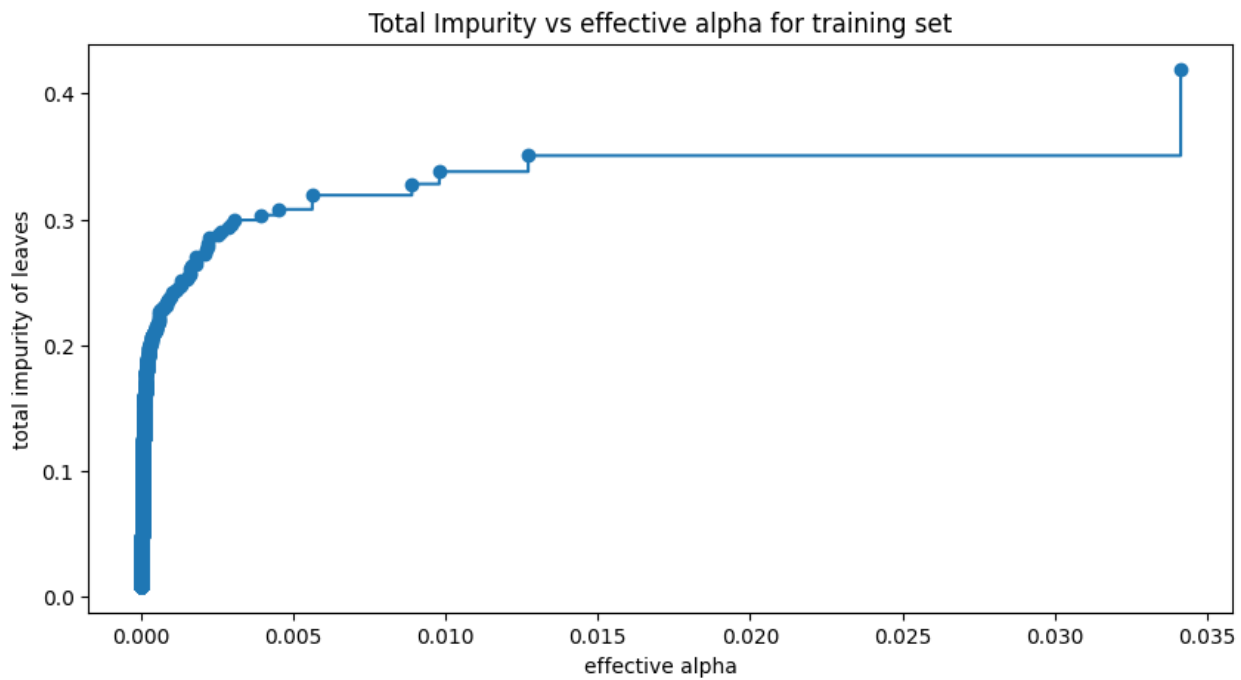


Figure 79: Total Impurity vs effective alpha for training set

Number of nodes in the last tree is: 1 with ccp_alpha: 0.08117914389136943

Decision Tree Pruning Outcome

After cost-complexity pruning, the final tree has only 1 node with an effective ccp_alpha of 0.0812.

Interpretation:

- The pruning process has simplified the tree maximally, indicating that only the most critical split (or no split at all) remains.
- This could happen if the tree is overly complex initially or the data does not support deeper splits without overfitting.

Business Insight:

A very simple tree may lack granularity to capture subtle patterns but ensures robustness and avoids overfitting. Further analysis may be needed to adjust alpha or explore other tree-based models for more detailed insights on cancellations.

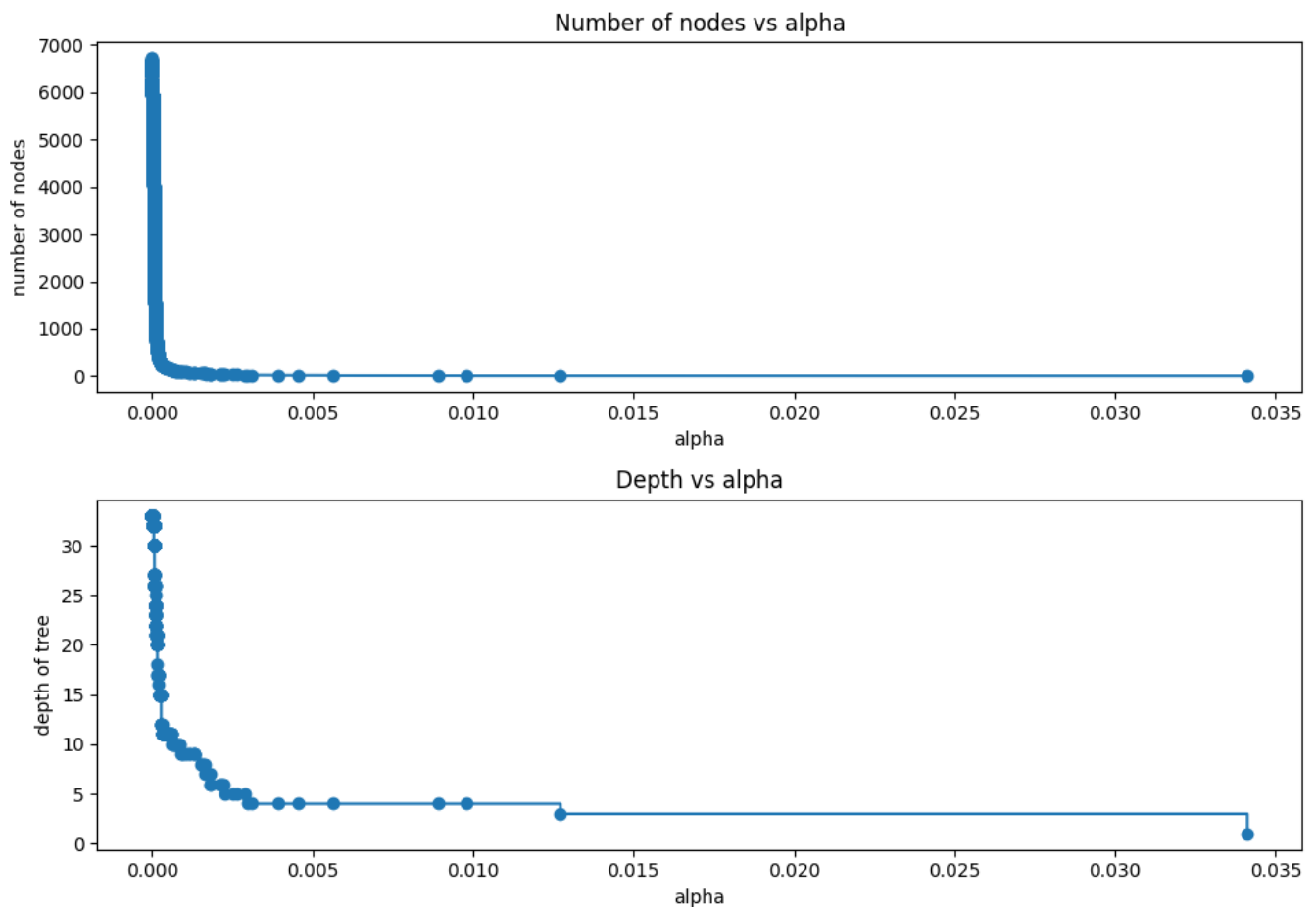


Figure 80: Number of nodes vs alpha and Depth vs alpha

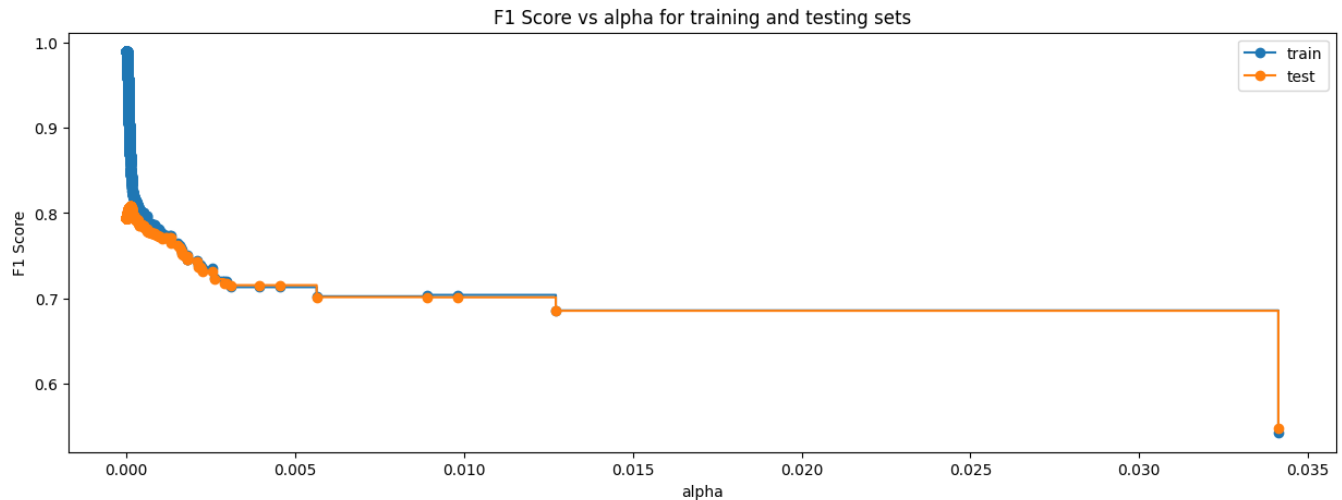


Figure 81: F1 Score vs alpha for training and testing sets

```
DecisionTreeClassifier(ccp_alpha=np.float64(0.0001226763315516701),
                      class_weight='balanced', random_state=1)
```

Figure 82: Best Model

`ccp_alpha = 0.0001227:`

This small alpha value indicates minimal pruning, allowing the tree to retain most splits while controlling overfitting.

`class_weight = 'balanced':`

Adjusts for imbalanced classes (canceled vs. not canceled bookings) to ensure the model gives equal attention to both outcomes.

`random_state = 1:`

Ensures reproducibility of results across runs.

Business Insight:

The model is configured to capture cancellation patterns effectively while mitigating bias from class imbalance, providing reliable guidance for operational decision-making.

Checking performance on training set

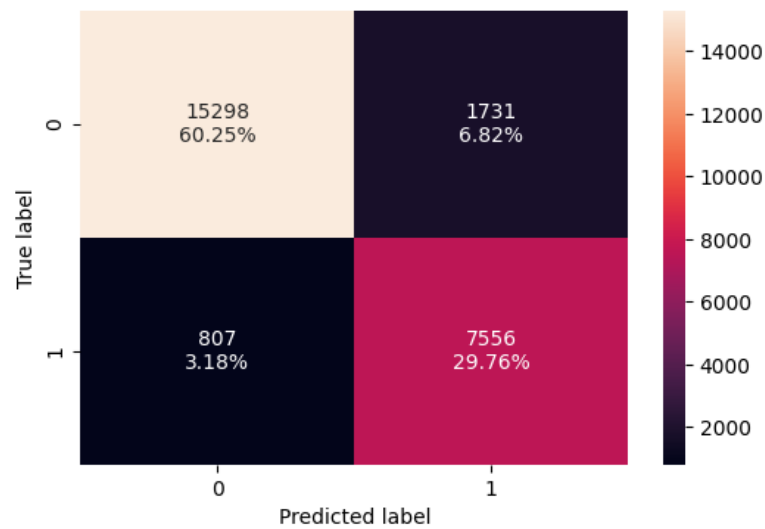


Figure 83: DT - CM 5

	Accuracy	Recall	Precision	F1
0	0.90005	0.90350	0.81361	0.85620

Figure 84: TPM 4 - Training

Observation:

- The model shows strong performance on the training data, correctly identifying most cancellations.
- High recall (~90.4%) indicates it captures the majority of canceled bookings.
- Precision (~81.4%) is slightly lower, indicating some false positives.
- F1 score (~85.6%) reflects a good balance between recall and precision.

Business Insight:

This tree-based model can effectively support operational decisions to reduce booking cancellations while maintaining high detection accuracy.

Checking performance on test set

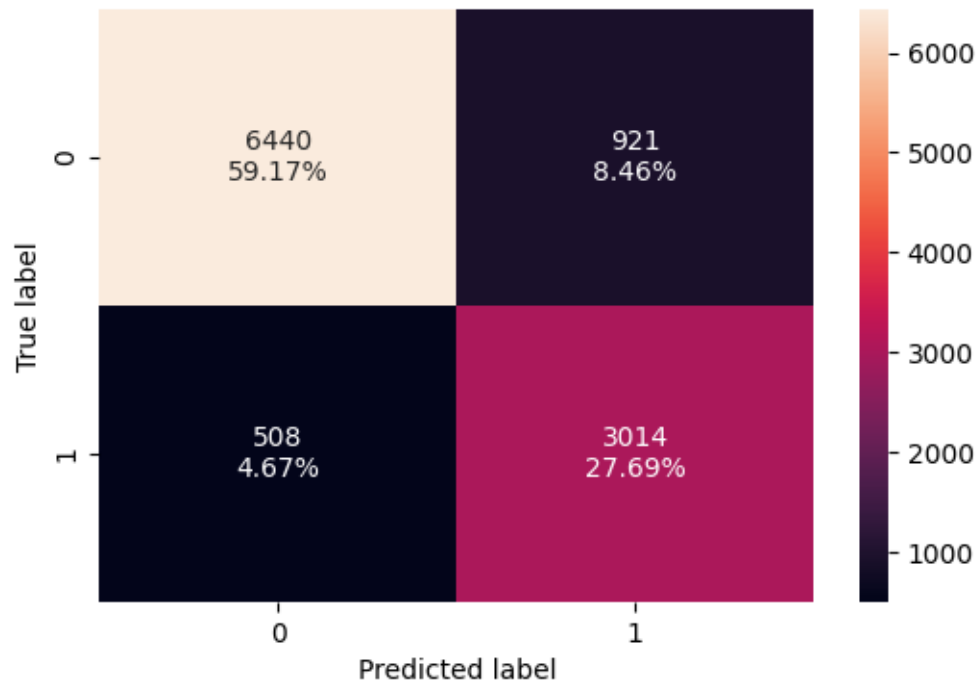


Figure 85: DT - CM 6

	Accuracy	Recall	Precision	F1
0	0.86869	0.85576	0.76595	0.80837

Figure 86: TPM 5 - Test

Observation:

- The model performs well on unseen data, showing strong generalization.
- Recall (~85.6%) indicates most cancellations are correctly identified, helping proactively manage bookings.
- Precision (76.6%) reflects some false positives, but the F1 score (80.8%) indicates a balanced performance.

Business Insight:

The model can be used operationally to anticipate cancellations and optimize resource allocation, while maintaining robust predictive accuracy.

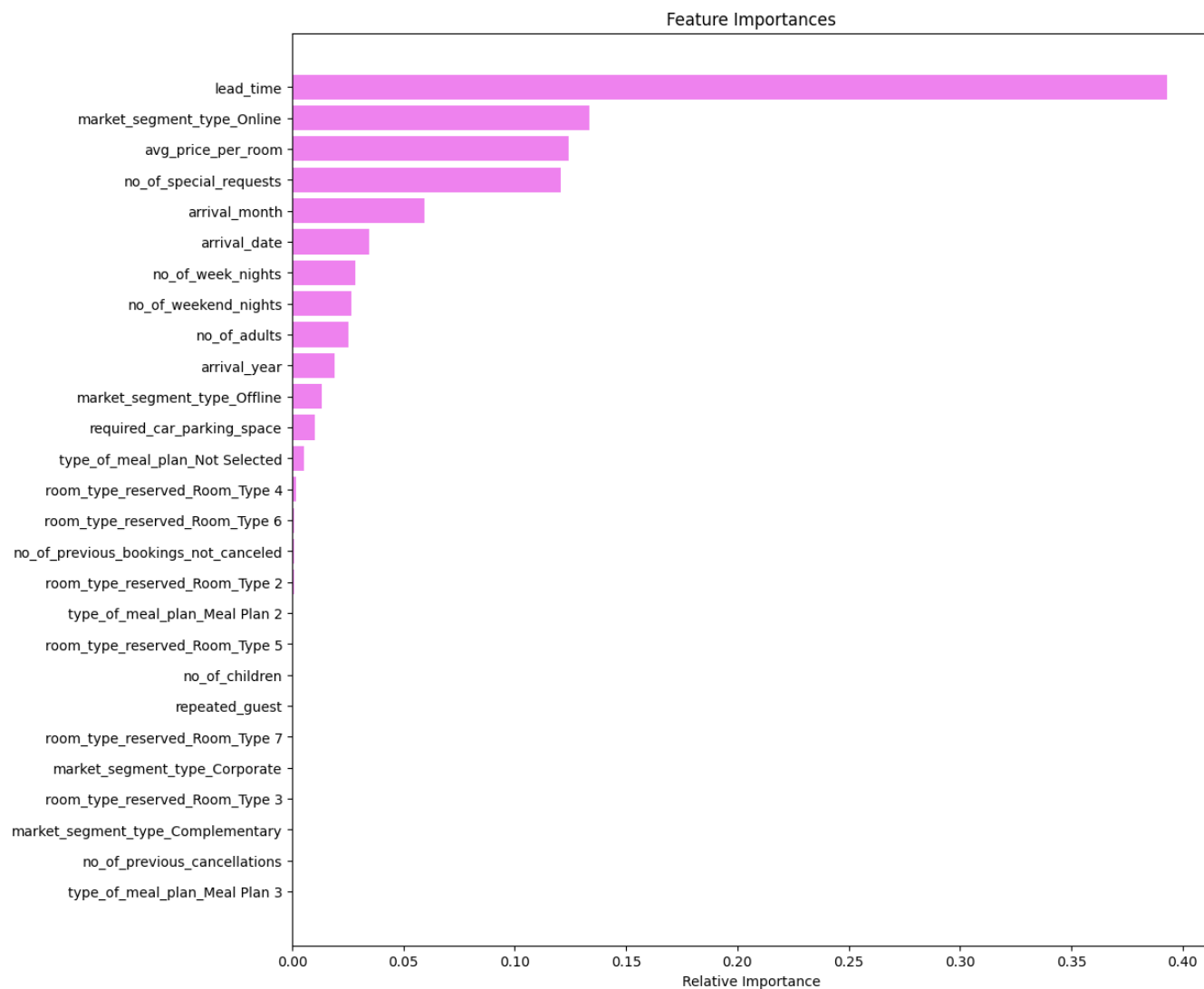


Figure 88: Feature Importance 3

Model Performance Comparison and Conclusions

Training performance comparison

Training performance comparison:

	Decision Tree sklearn	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.99421	0.83101	0.90005
Recall	0.98661	0.78620	0.90350
Precision	0.99578	0.72428	0.81361
F1	0.99117	0.75397	0.85620

Figure 89: Training performance comparison

Observation:

- Default decision tree shows extremely high training performance but may be overfitting.
- Pre-pruning reduces overfitting but decreases accuracy and recall.
- Post-pruning strikes a better balance, maintaining high recall and improved F1 score while controlling overfitting.

Business Insight:

Post-pruned trees provide robust predictions, capturing cancellations effectively while avoiding overly complex models.

Test Set performance comparison

Test set performance comparison:

	Decision Tree with class_weight	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.87108	0.83497	0.86869
Recall	0.81034	0.78336	0.85576
Precision	0.79521	0.72758	0.76595
F1	0.80270	0.75444	0.80837

Figure 90: Test Set performance comparison

Observation:

- Class-weighted tree performs well, balancing precision and recall.
- Pre-pruned tree has slightly lower accuracy and F1, reflecting a more conservative model.
- Post-pruned tree improves recall and F1, making it effective for capturing cancellations while avoiding overfitting.

Business Insight:

Post-pruned decision trees are recommended for operational use, providing robust predictions and actionable insights for managing cancellations efficiently.

Actionable Insights & Recommendations

Actionable Insights

Booking Patterns & Guest Behaviour

- Most bookings are for 2 adults (72%) with no children (92.6%) and typically include weekdays (1–3 nights).
- Popular meal plan is Meal Plan 1 (breakfast only, 76.7%).
- Online market segment (64%) dominates bookings; “Offline” is also significant (29%).
- Lead time is a key driver: majority of guests book within 4 months of arrival; longer lead times correlate with higher room prices and slightly higher cancellations.
- Repeated guests (~2.6%) tend to have fewer adults/children and fewer special requests.

Cancellations

- Overall, 32.8% of bookings are canceled.
- Cancellation likelihood increases with longer lead times, more adults/children, and more special requests.
- Market segments differ:
- Online bookings have higher cancellations (8,475 of 23,214).
- Offline and Corporate have fewer cancellations.
- Complementary bookings rarely cancel.

Model Insights

- Logistic Regression & Decision Trees show that lead time, number of adults/children, special requests, and market segment are the most important predictors.
- Post-pruned Decision Trees provide a balanced performance (F1 ~80.8% on test data) while avoiding overfitting.
- Adjusting classification thresholds can increase recall for cancellations, enabling more proactive management.

Recommendations

Proactive Cancellation Management

- Monitor high-risk bookings (long lead times, multiple adults/children, multiple special requests).
- Consider automated reminders or incentives to reduce cancellations for these bookings.
- Use market segment data to target interventions (e.g., Online vs. Corporate).

Revenue & Pricing Strategy

- Adjust pricing for longer stays and bookings with more guests, as these are correlated with higher room prices.
- Introduce flexible pricing or packages for early bookings to encourage commitment and reduce cancellations.

Operational Planning

- Allocate resources (staff, rooms) based on seasonal demand patterns, with peaks in August–October.
- Track special requests closely, as higher numbers correlate with cancellations and operational complexity.

Model Deployment & Monitoring

- Use post-pruned decision tree model to flag potential cancellations.
- Adjust thresholds to maximize recall if reducing cancellations is the priority.
- Regularly retrain the model with new data to capture changing booking behaviors.

Customer Retention & Loyalty Programs

- Focus on first-time and high-risk guests to reduce cancellations.
- Encourage repeat bookings with discounts, loyalty points, or personalized offers.